

Supplementary File for “Deep Learning-based Search Space Reconstruction Empowers Evolution Algorithm for Optimization”

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Zhenyu Lei, and Shangce Gao, *Senior Member, IEEE*

THIS part provides the supplementary materials for the body of “Deep Learning-based Search Space Reconstruction Empowers Evolution Algorithm for Optimization”. As shown in Table ??, the parameter settings of algorithms are configured as the recommended values according to the references. Tables ?? —?? present the consolidated experimental results of DLES and state-of-the-art competitive algorithms on CEC 2014 benchmark functions across dimensions 10, 30, 50, and 100. Similarly, Tables ?? —?? showcase the summarized experimental outcomes of DLES and leading competitive algorithms on CEC 2018 benchmark functions in the same dimensions. Additionally, Table ?? and Table ?? exhibit the consolidated experimental findings of DLES and state-of-the-art competitive algorithms on CEC 2022 benchmark functions in dimensions 10 and 20, respectively.

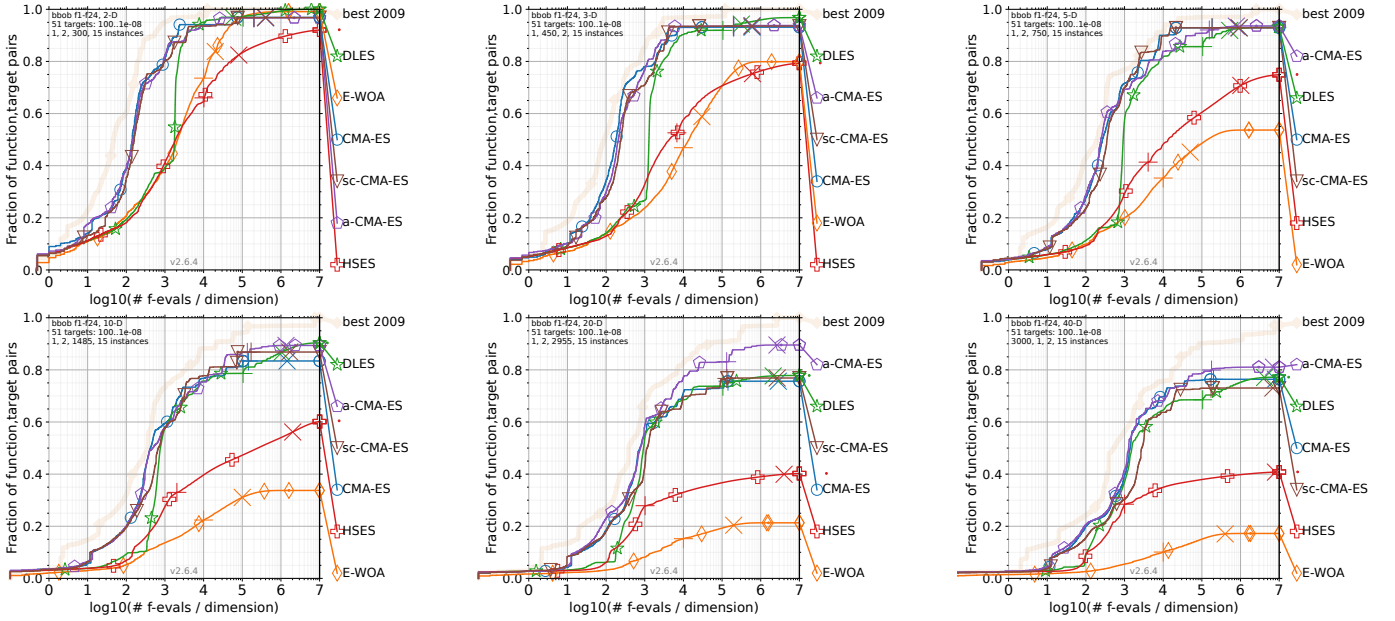


Fig. S.I: Overview of experimental results on the BBOB dataset

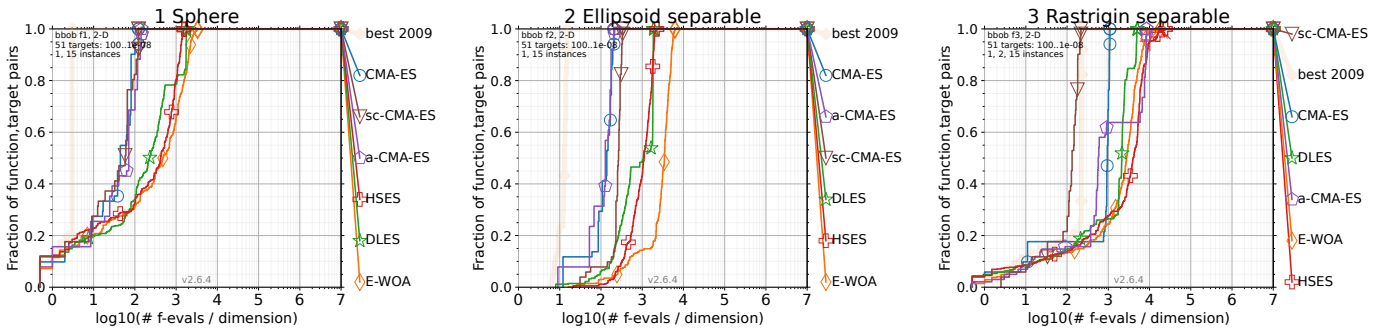


Fig. S.II: F1 to F3 in D=2.

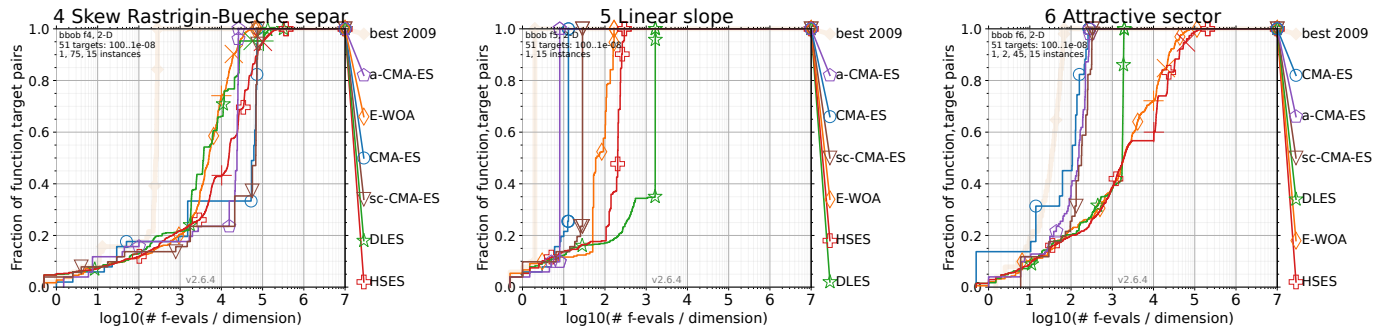


Fig. S.III: F4 to F6 in D=2.

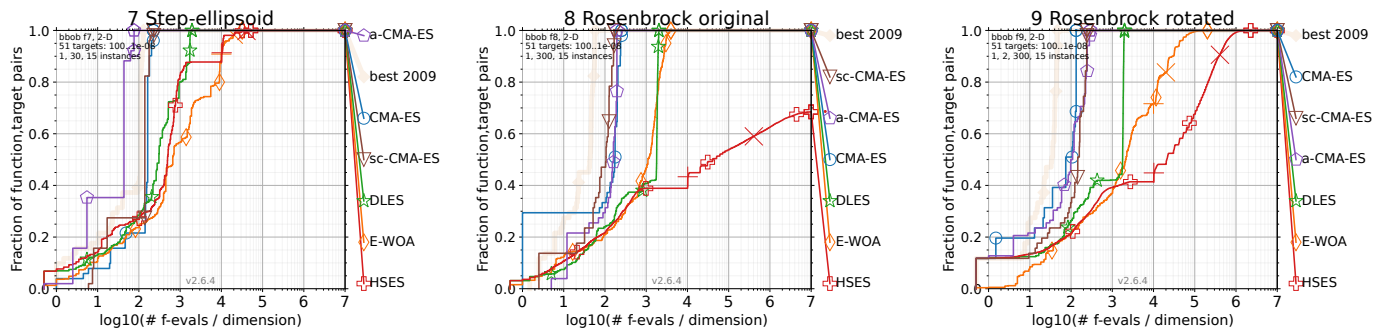


Fig. S.IV: F7 to F9 in D=2.

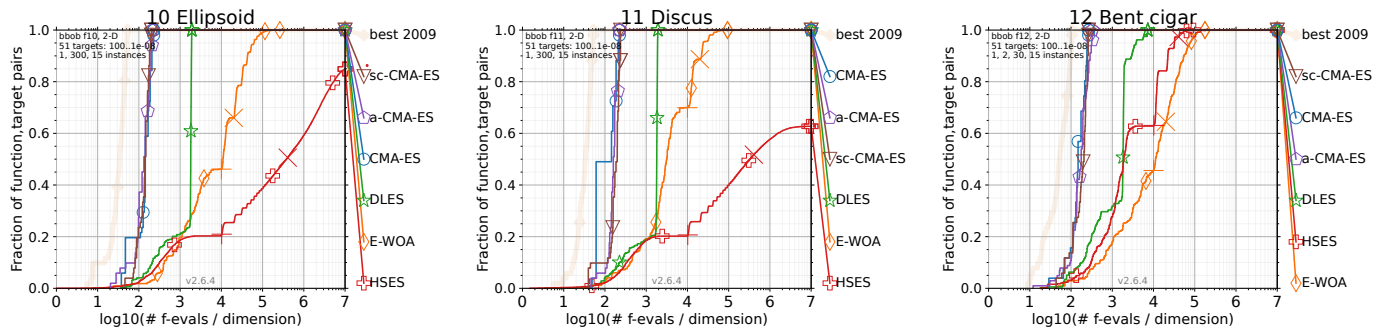


Fig. S.V: F10 to F12 in D=2.

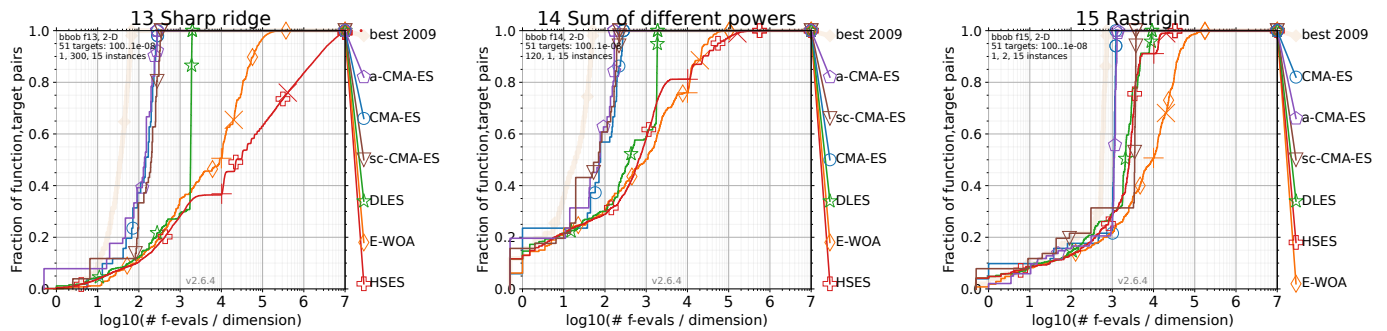


Fig. S.VI: F13 to F15 in D=2.

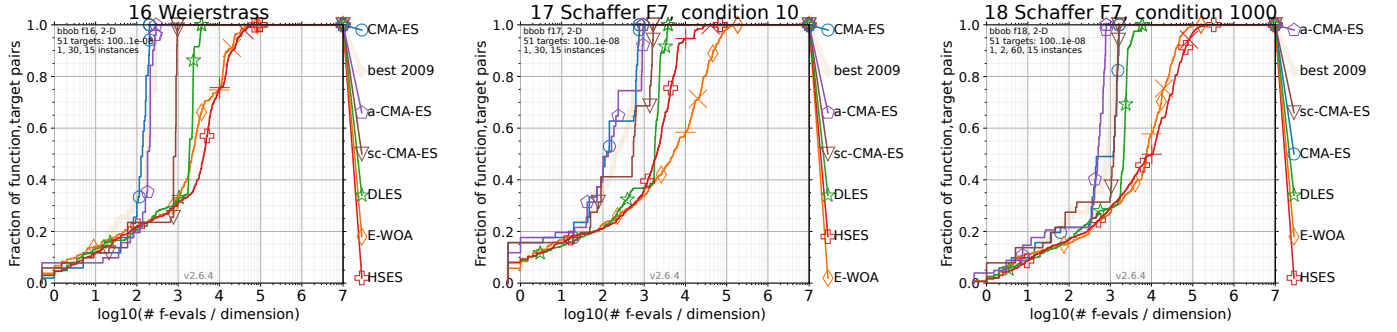


Fig. S.VII: F16 to F18 in D=2.

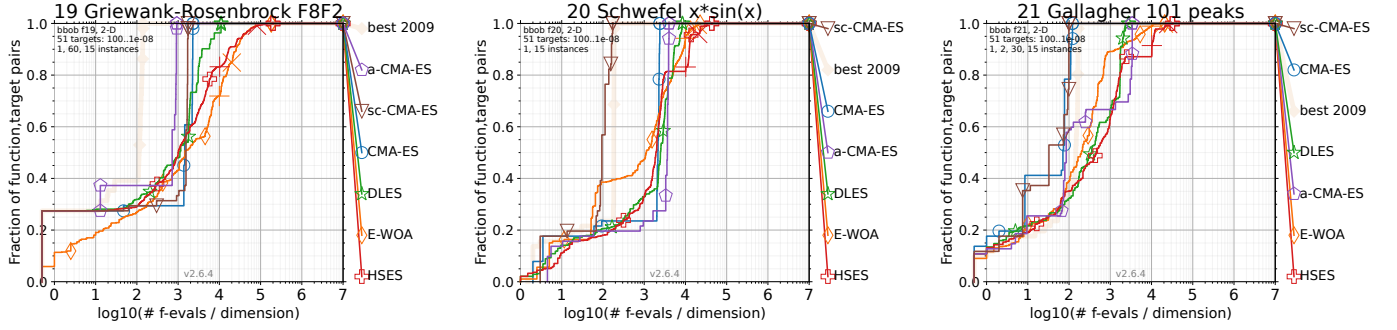


Fig. S.VIII: F19 to F21 in D=2.

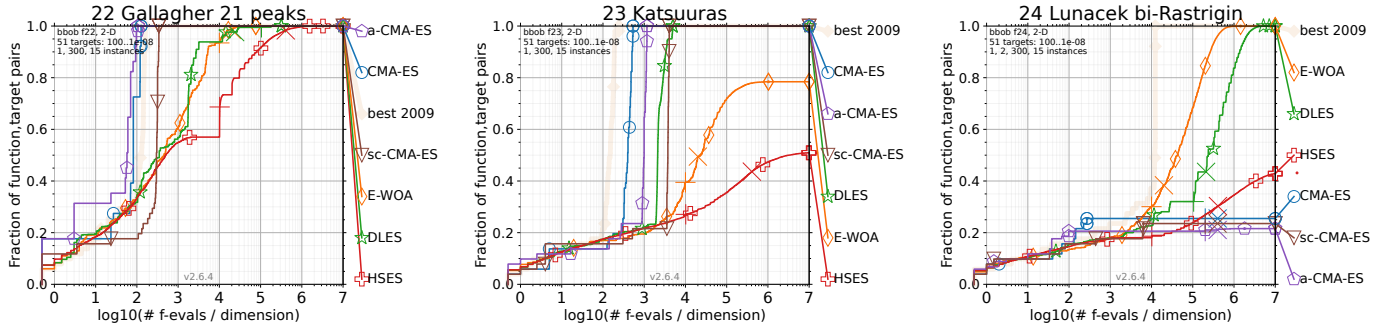


Fig. S.IX: F22 to F24 in D=2.

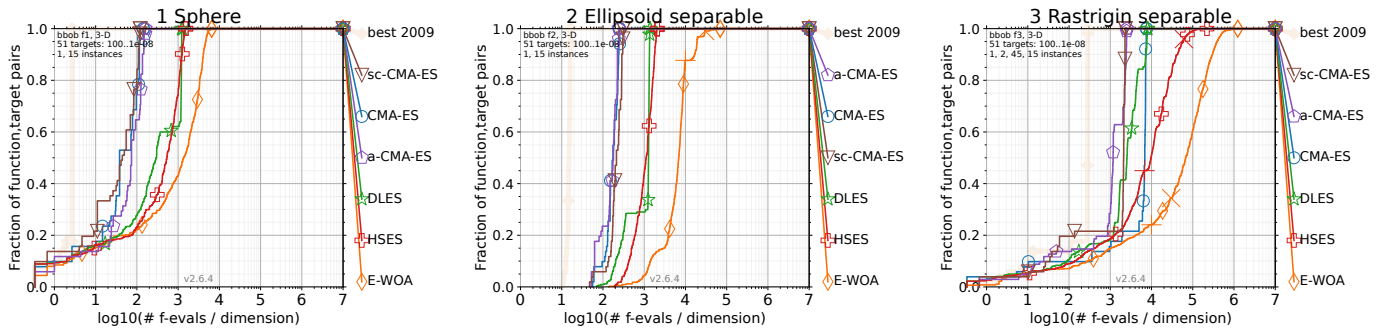


Fig. S.X: F1 to F3 in D=3.

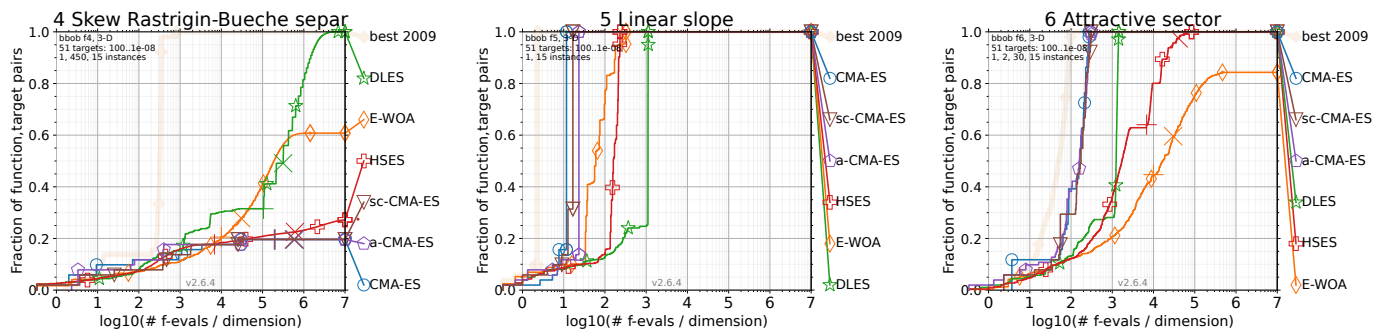


Fig. S.XI: F4 to F6 in D=3.

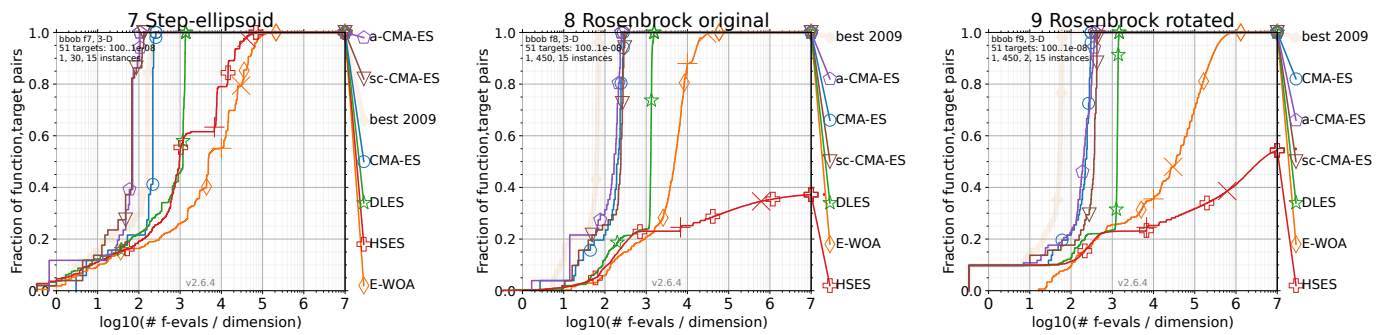


Fig. S.XII: F7 to F9 in D=3.

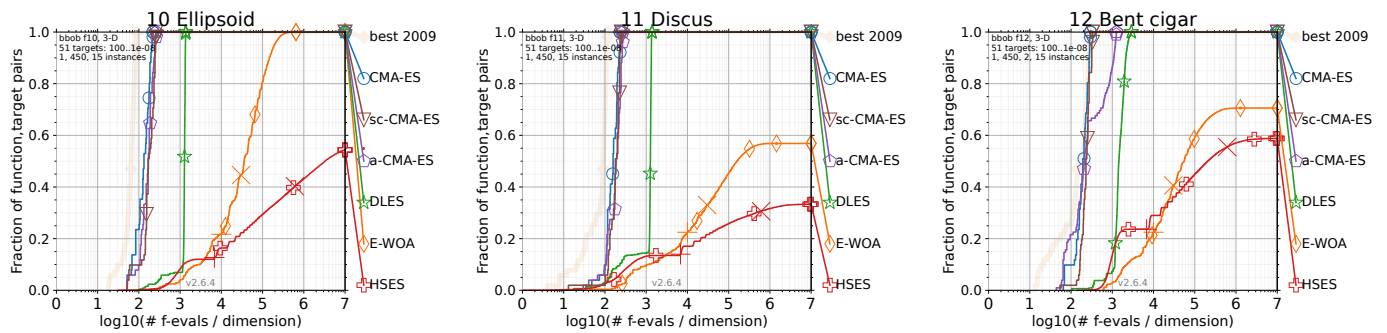


Fig. S.XIII: F10 to F12 in D=3.

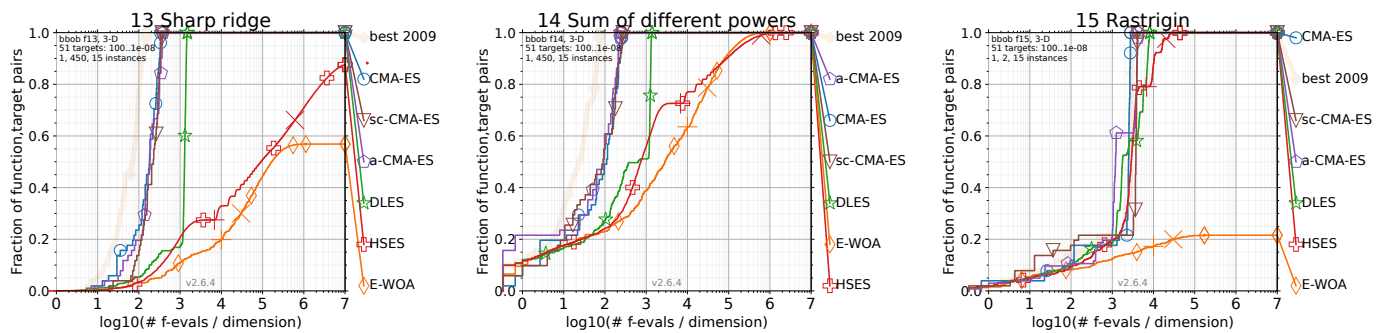


Fig. S.XIV: F13 to F15 in D=3.

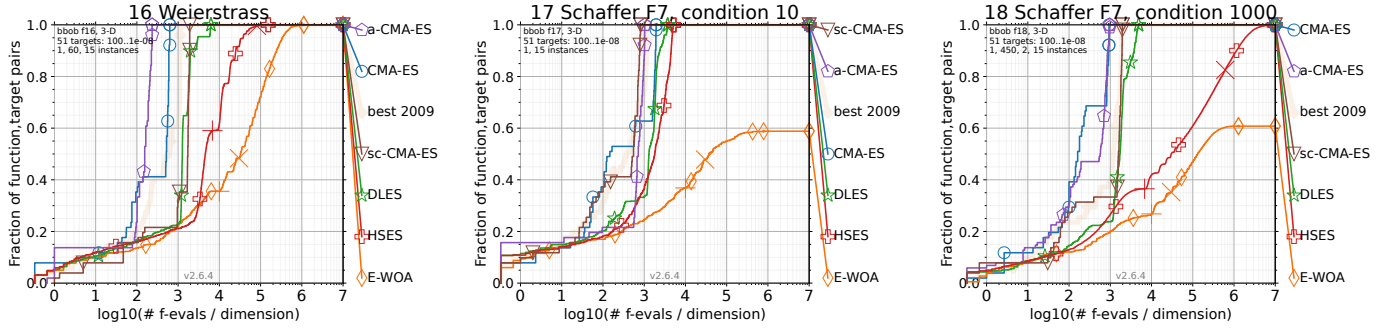


Fig. S.XV: F16 to F18 in D=3.

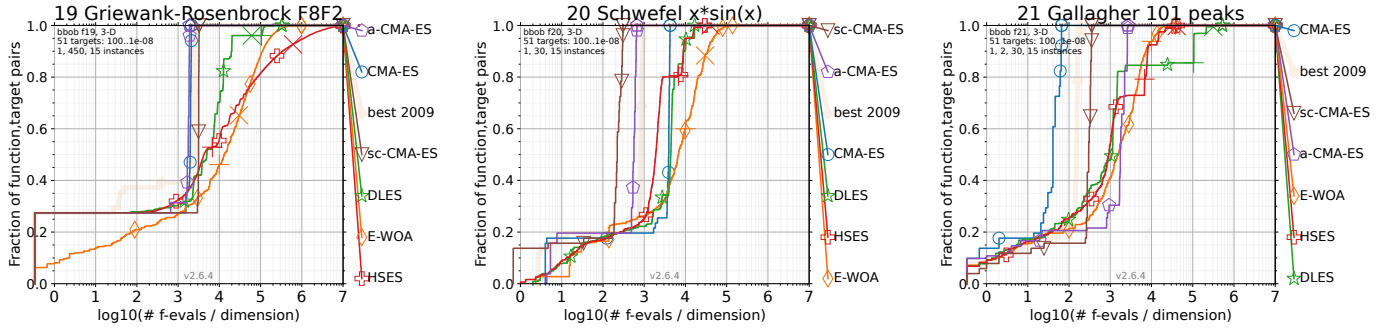


Fig. S.XVI: F19 to F21 in D=3.

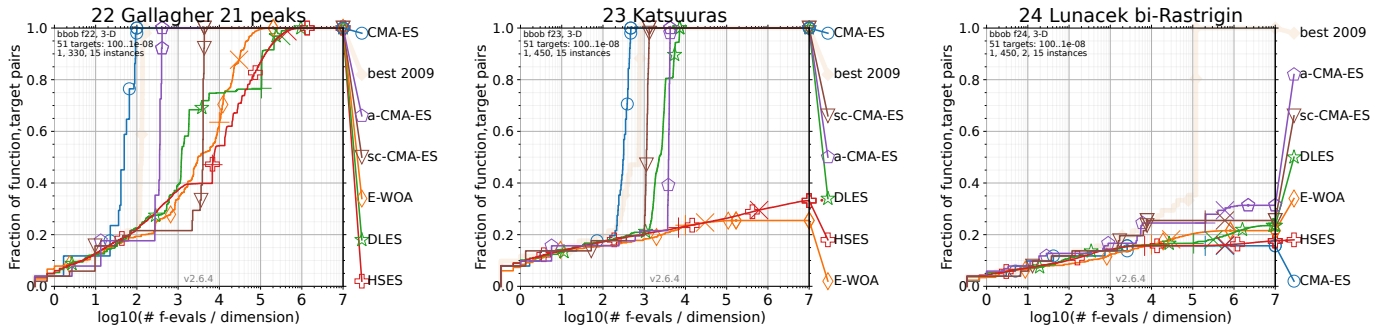


Fig. S.XVII: F22 to F24 in D=3.

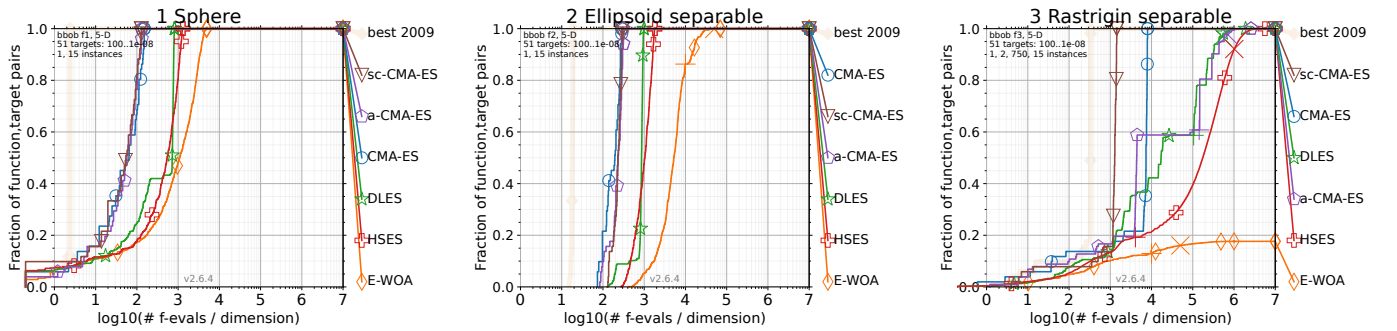


Fig. S.XVIII: F1 to F3 in D=5.

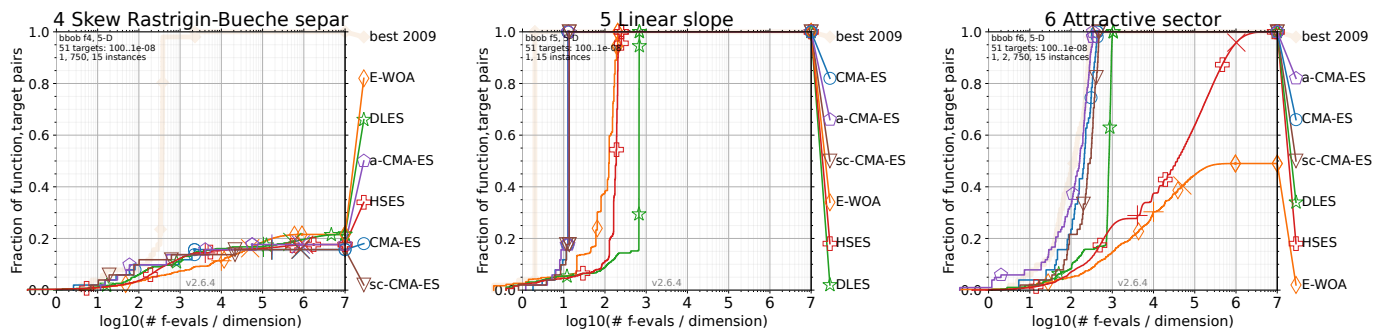


Fig. S.XIX: F4 to F6 in D=5.

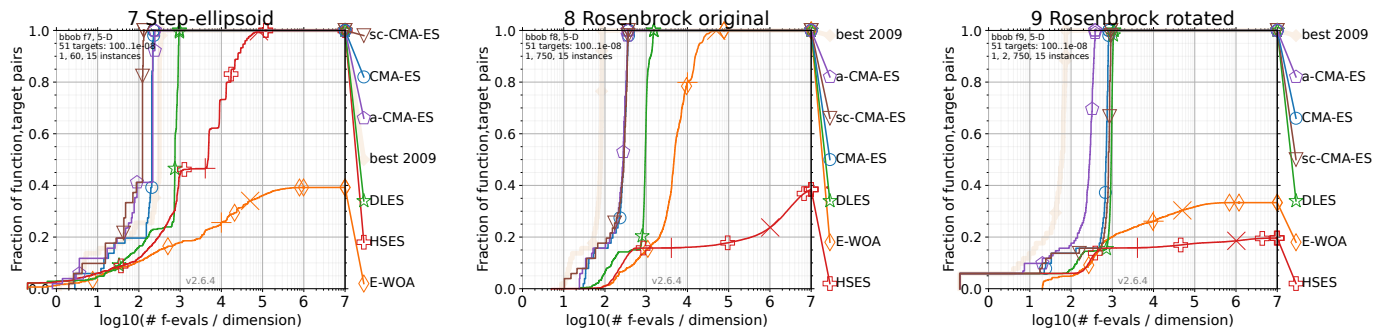


Fig. S.XX: F7 to F9 in D=5.

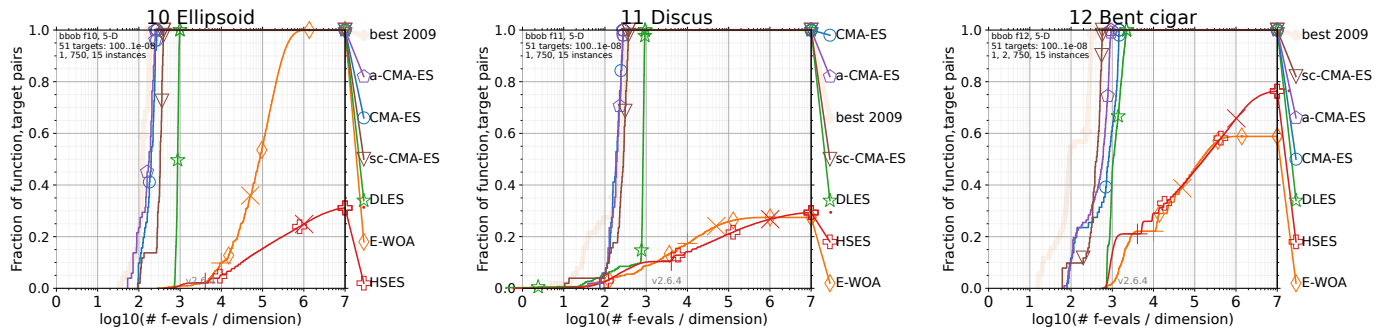


Fig. S.XXI: F10 to F12 in D=5.

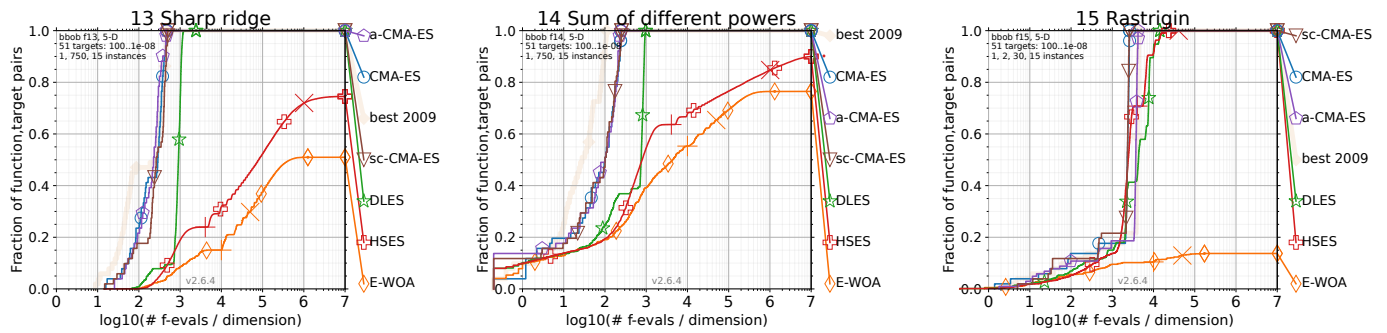


Fig. S.XXII: F13 to F15 in D=5.

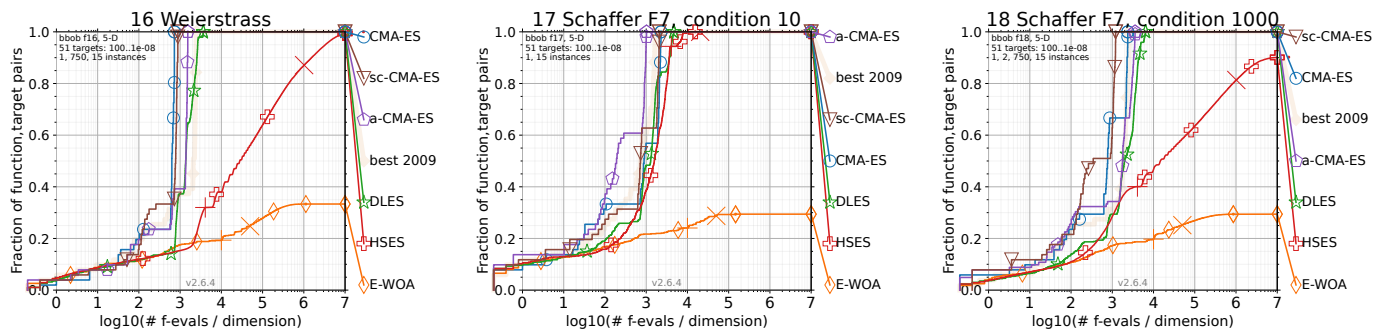


Fig. S.XXIII: F16 to F18 in D=5.

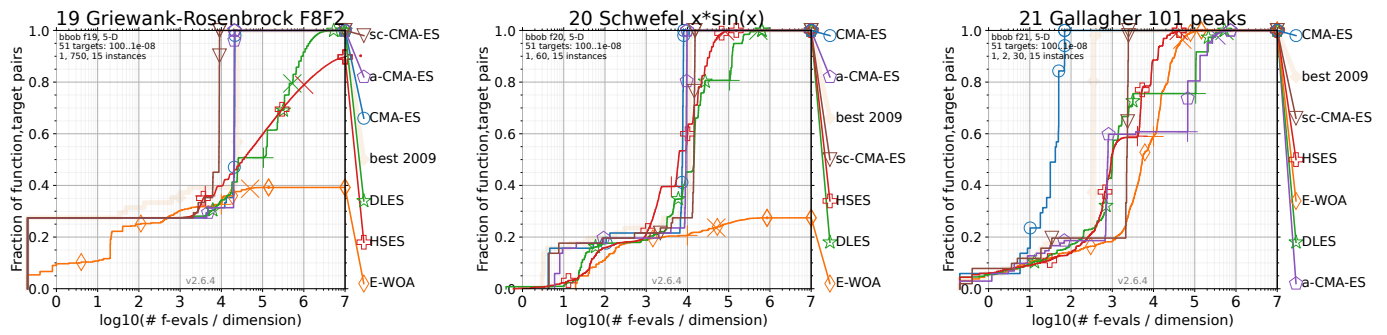


Fig. S.XXIV: F19 to F21 in D=5.

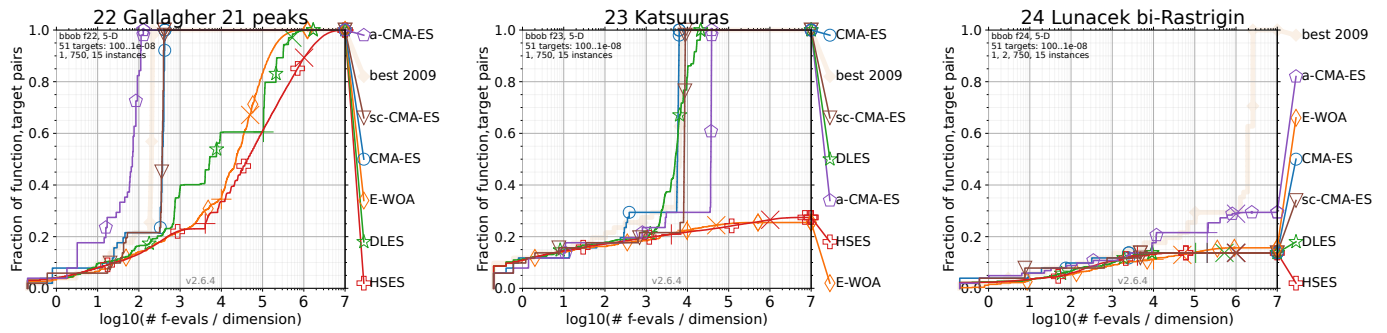


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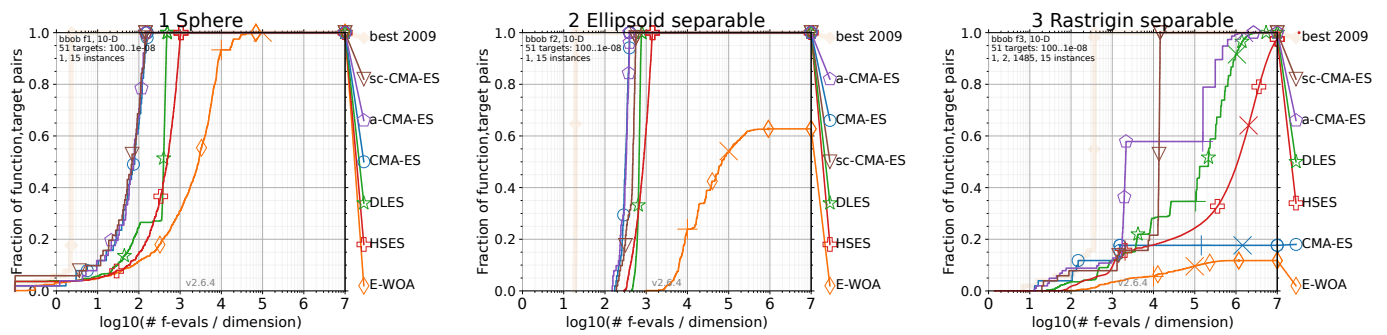


Fig. S.XXVI: F1 to F3 in D=10.

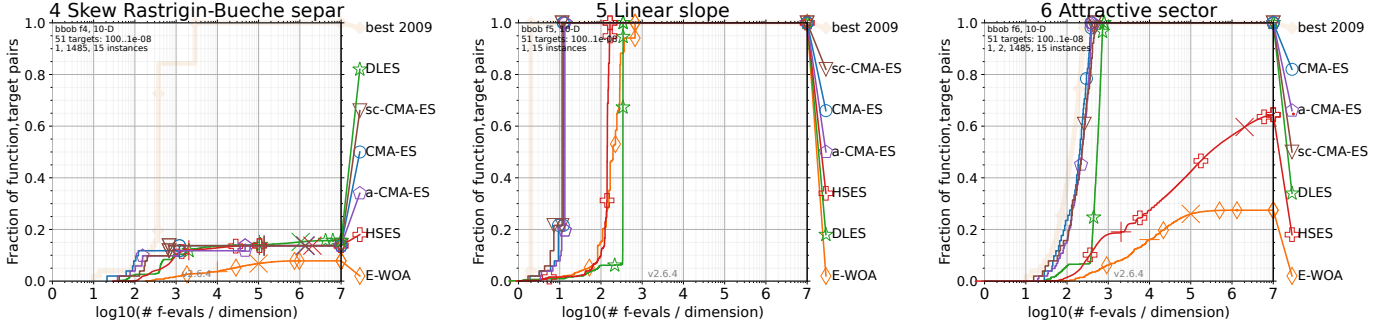


Fig. S.XXVII: F4 to F6 in D=10.

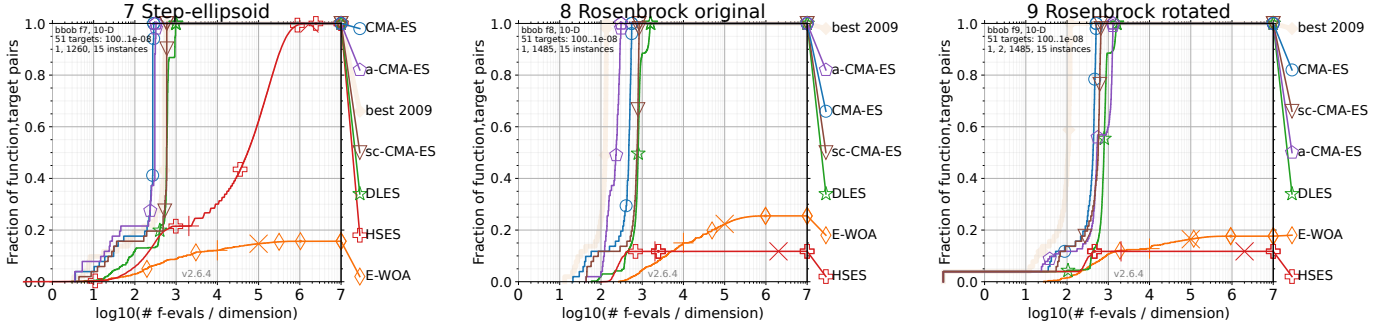


Fig. S.XXVIII: F7 to F9 in D=10.

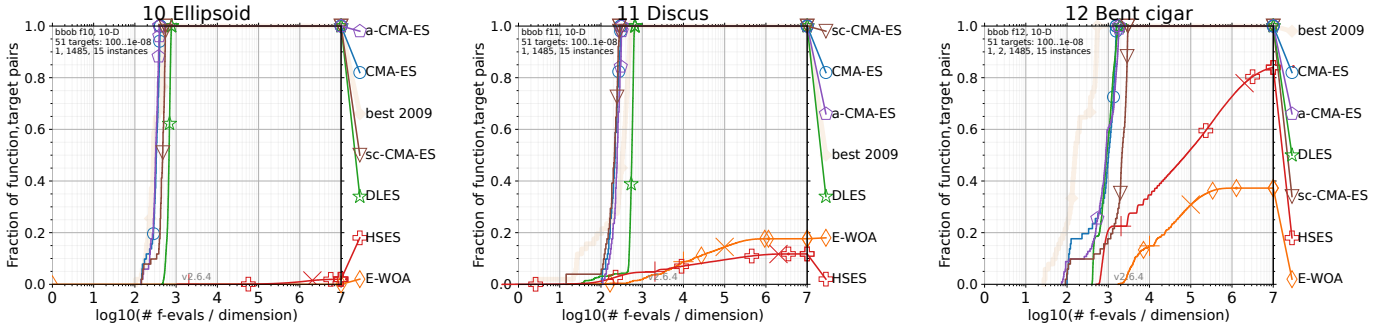


Fig. S.XXIX: F10 to F12 in D=10.

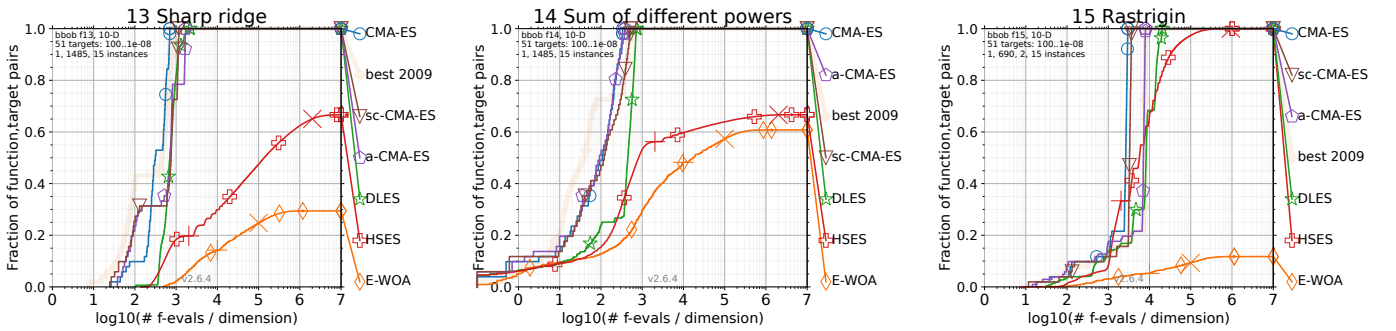


Fig. S.XXX: F13 to F15 in D=10.

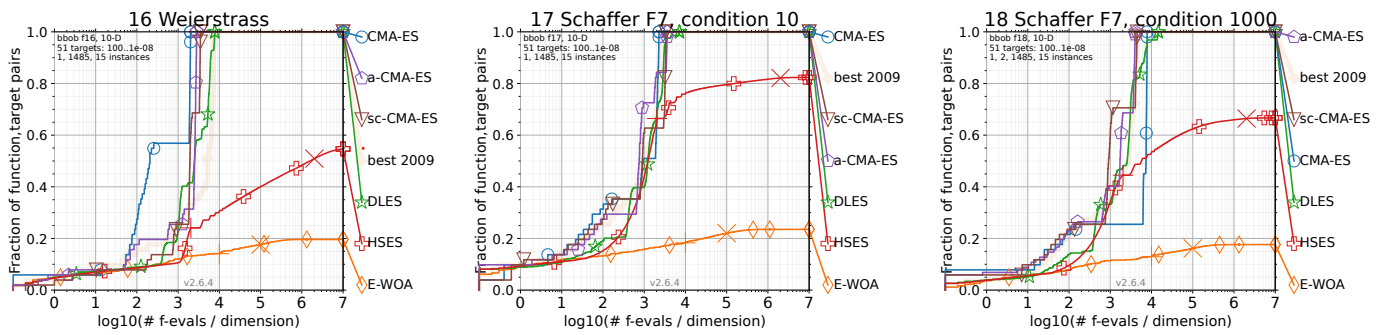


Fig. S.XXXI: F16 to F18 in D=10.

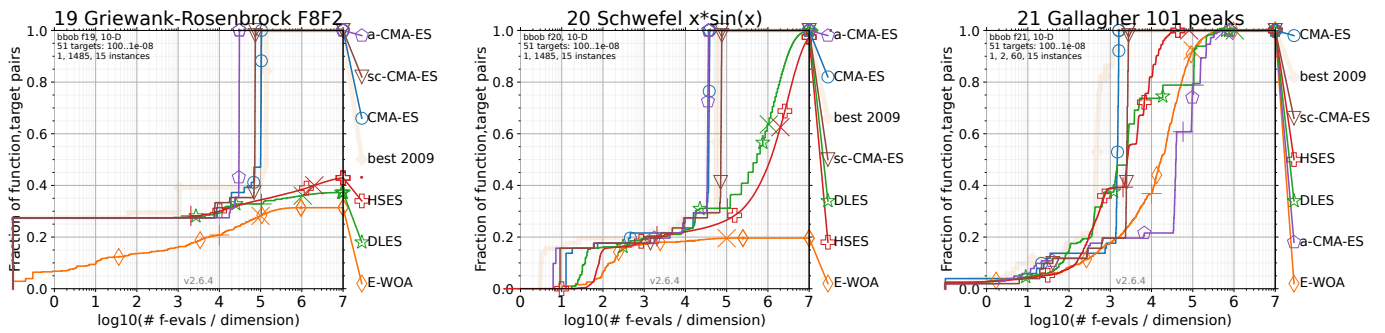


Fig. S.XXXII: F19 to F21 in D=10.

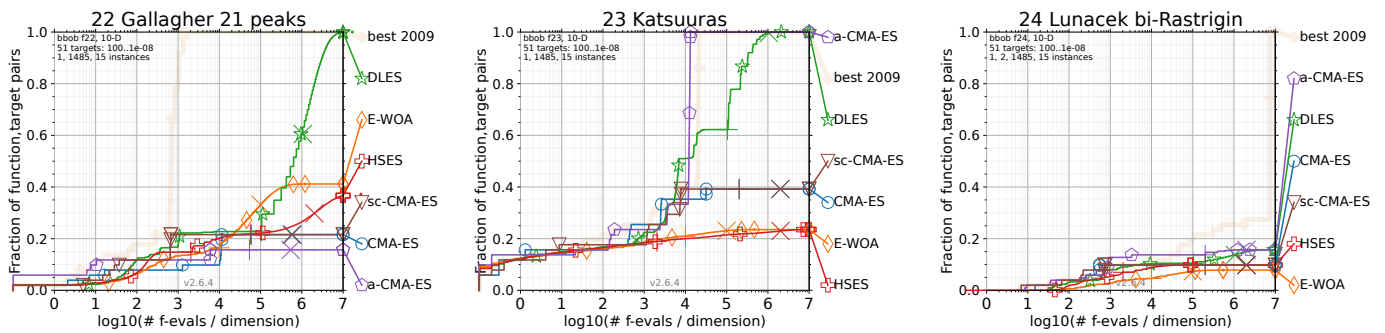


Fig. S.XXXIII: F22 to F24 in D=10.

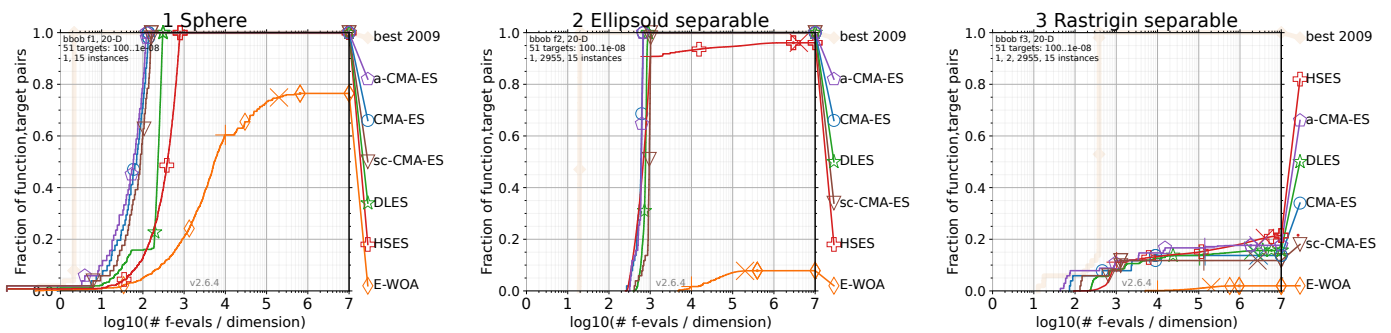


Fig. S.XXXIV: F1 to F3 in D=20.

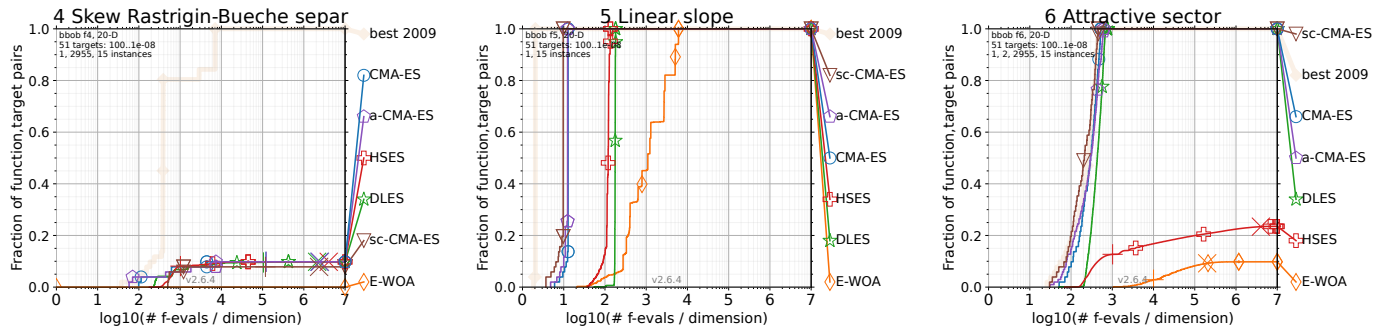


Fig. S.XXXV: F4 to F6 in D=20.

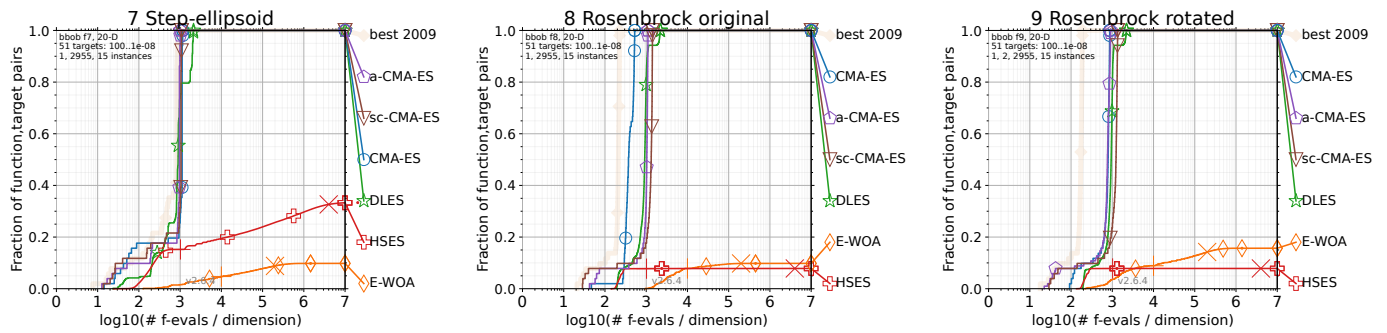


Fig. S.XXXVI: F7 to F9 in D=20.

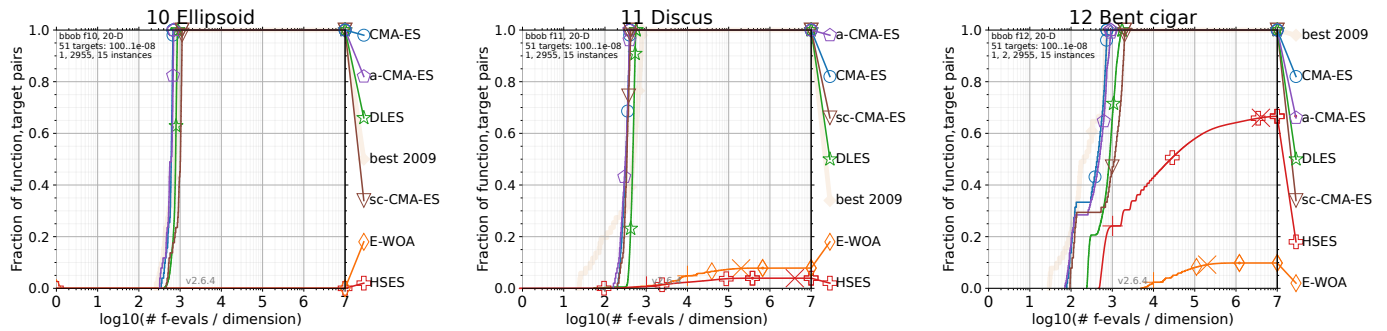


Fig. S.XXXVII: F10 to F12 in D=20.

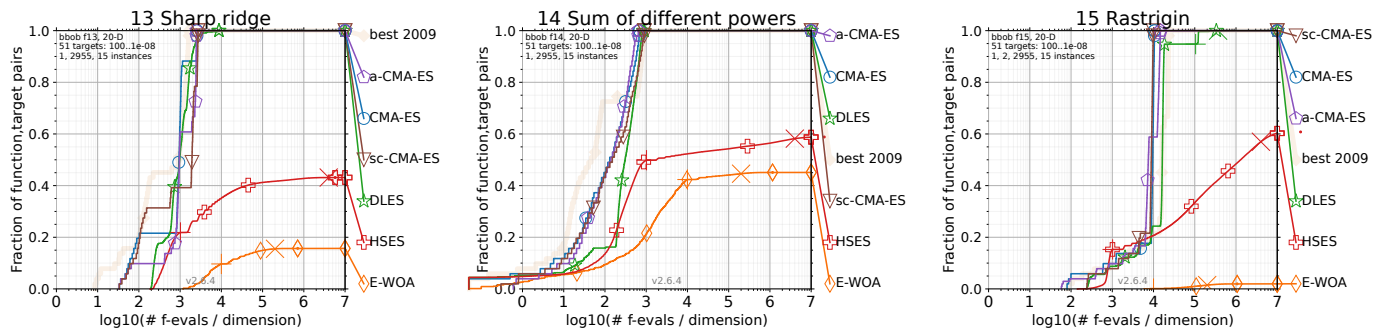


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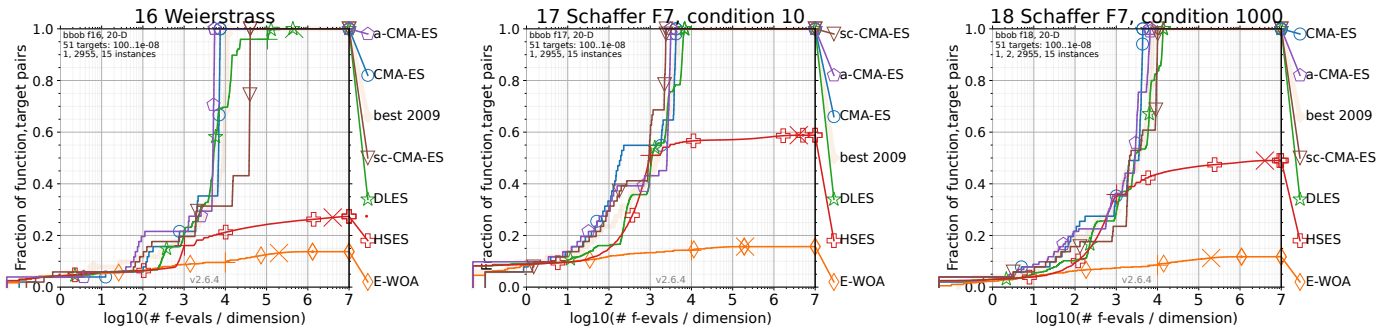


Fig. S.XXXIX: F16 to F18 in D=20.

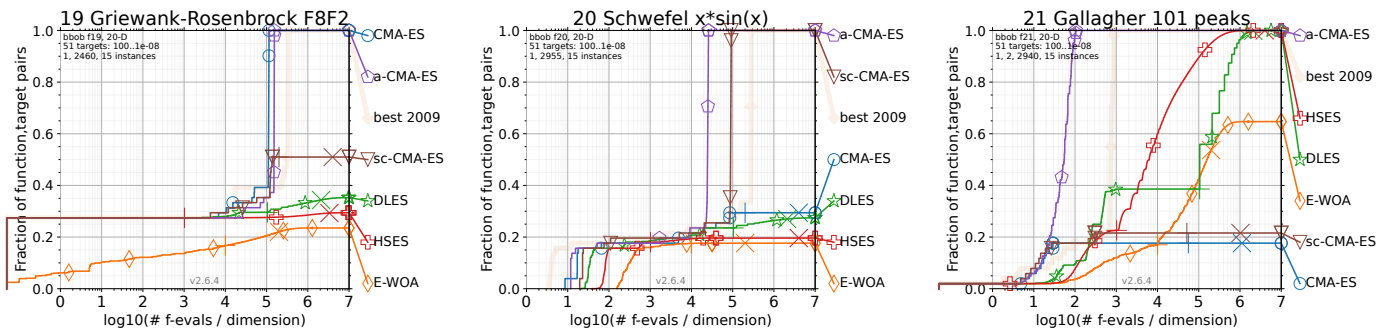


Fig. S.XL: F19 to F21 in D=20.

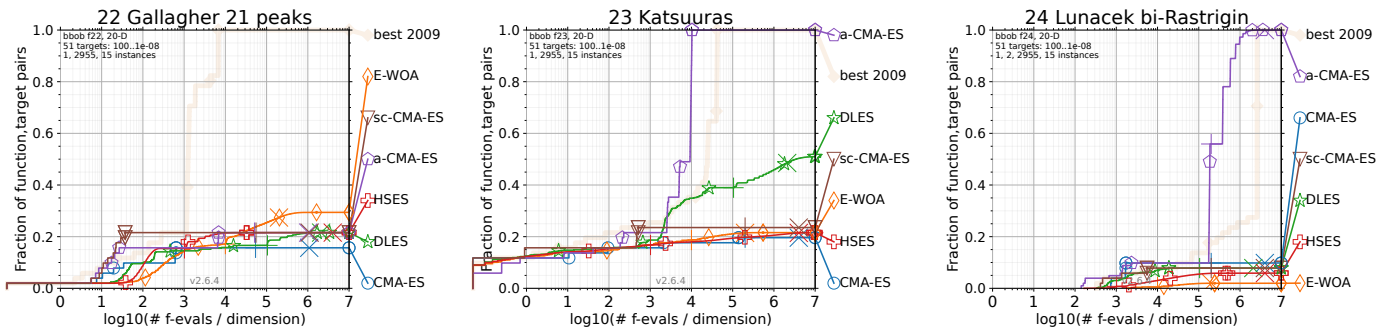


Fig. S.XLI: F22 to F24 in D=20.

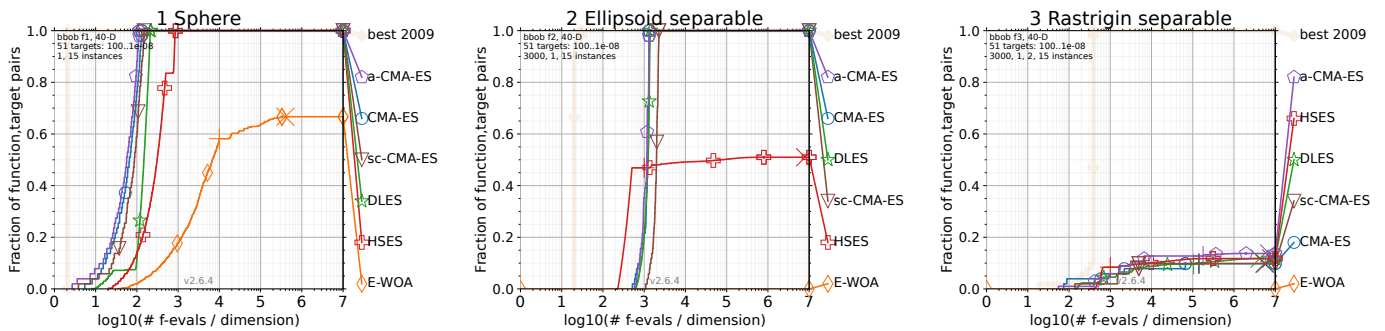


Fig. S.XLII: F1 to F3 in D=40.

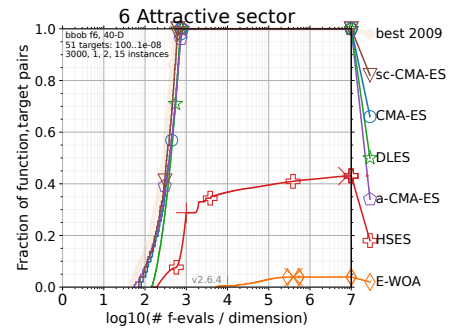
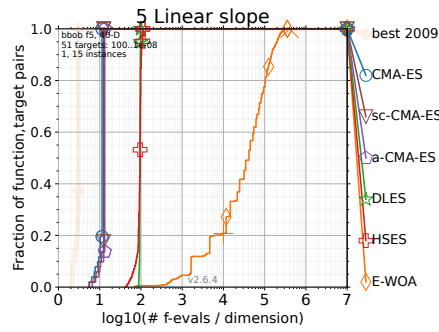
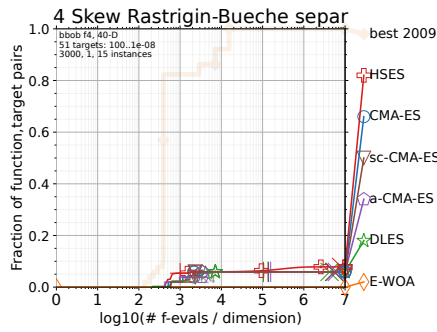


Fig. S.XLIII: F4 to F6 in D=40.

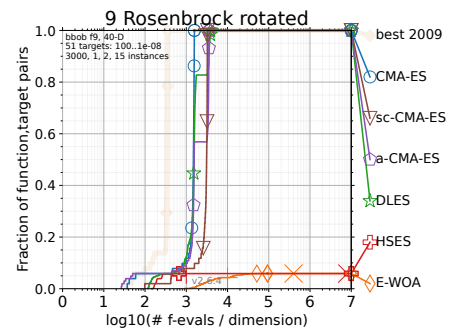
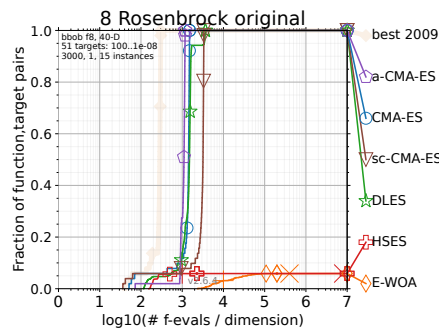
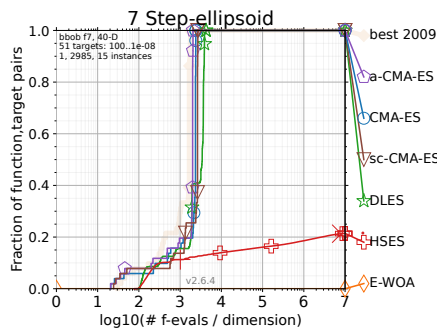


Fig. S.XLIV: F7 to F9 in D=40.

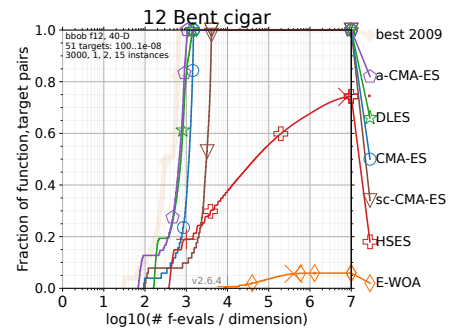
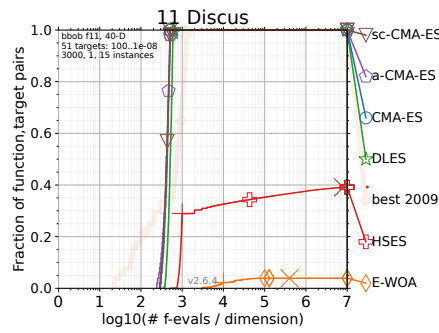
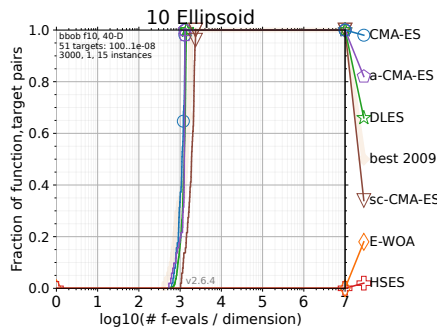


Fig. S.XLV: F10 to F12 in D=40.

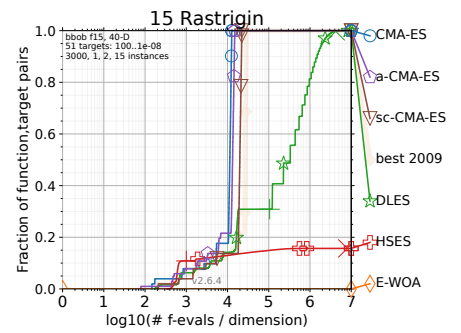
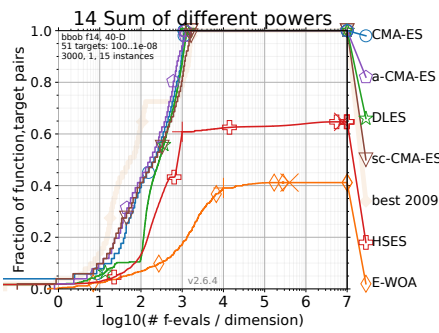
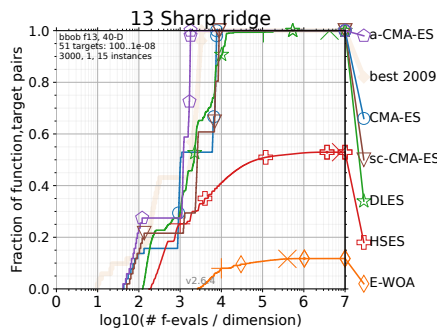


Fig. S.XLVI: F13 to F15 in D=40.

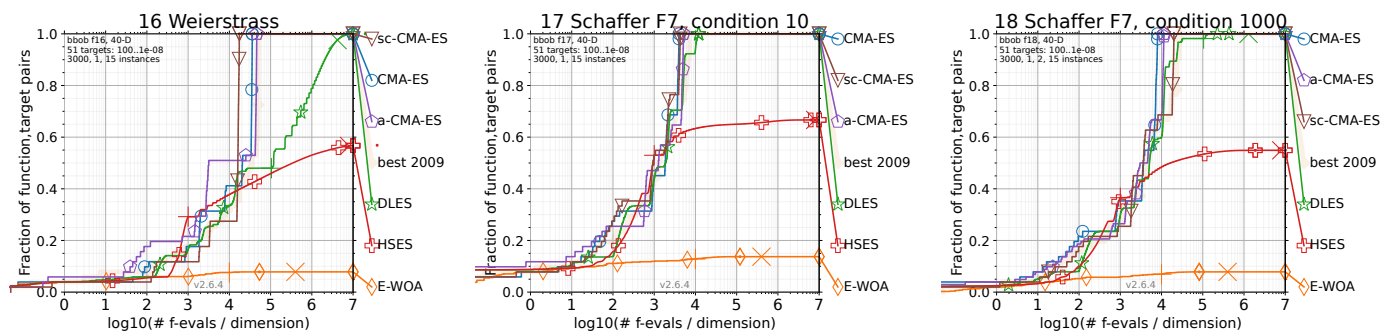


Fig. S.XLVII: F16 to F18 in D=40.

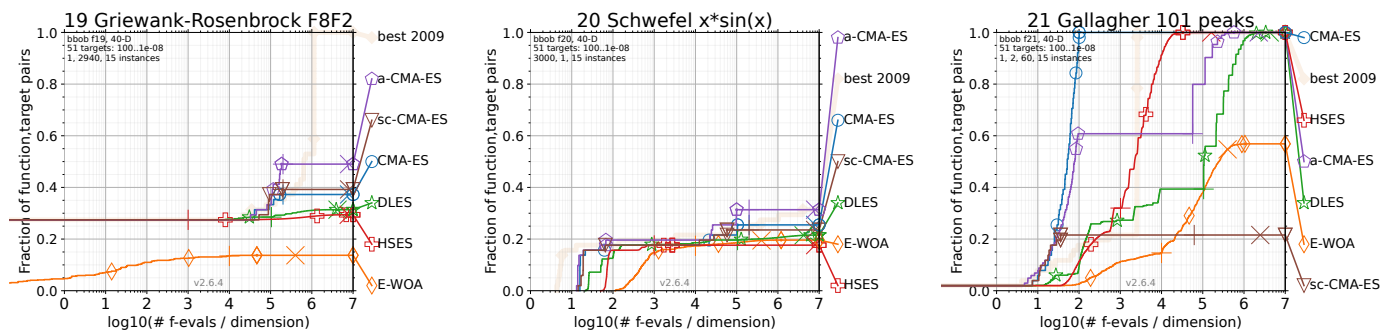


Fig. S.XLVIII: F19 to F21 in D=40.

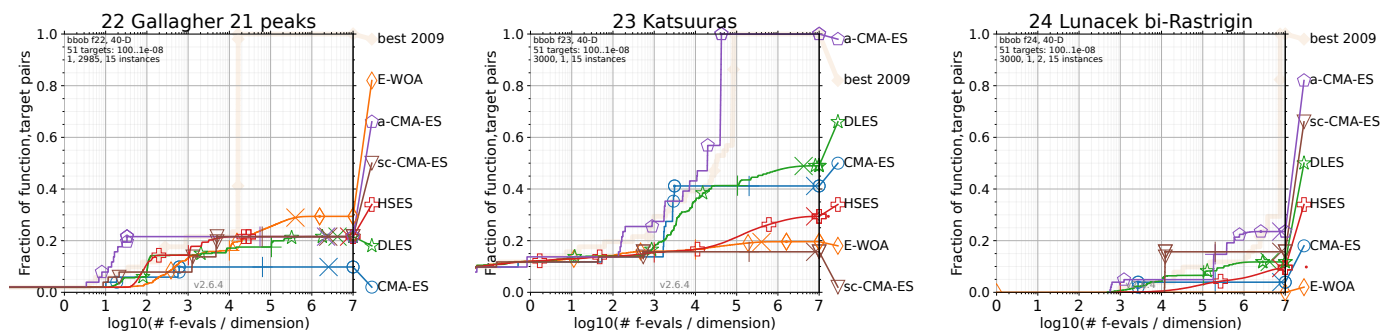


Fig. S.XLIX: F22 to F24 in D=40.

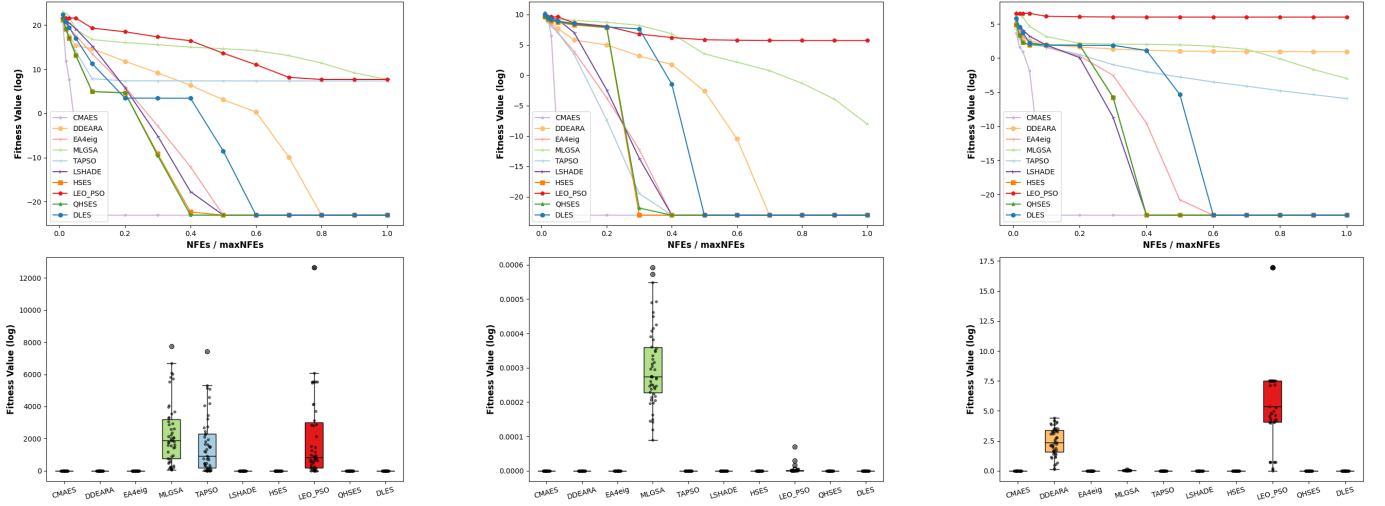


Fig. S.L: F1 to F3 in D=10.

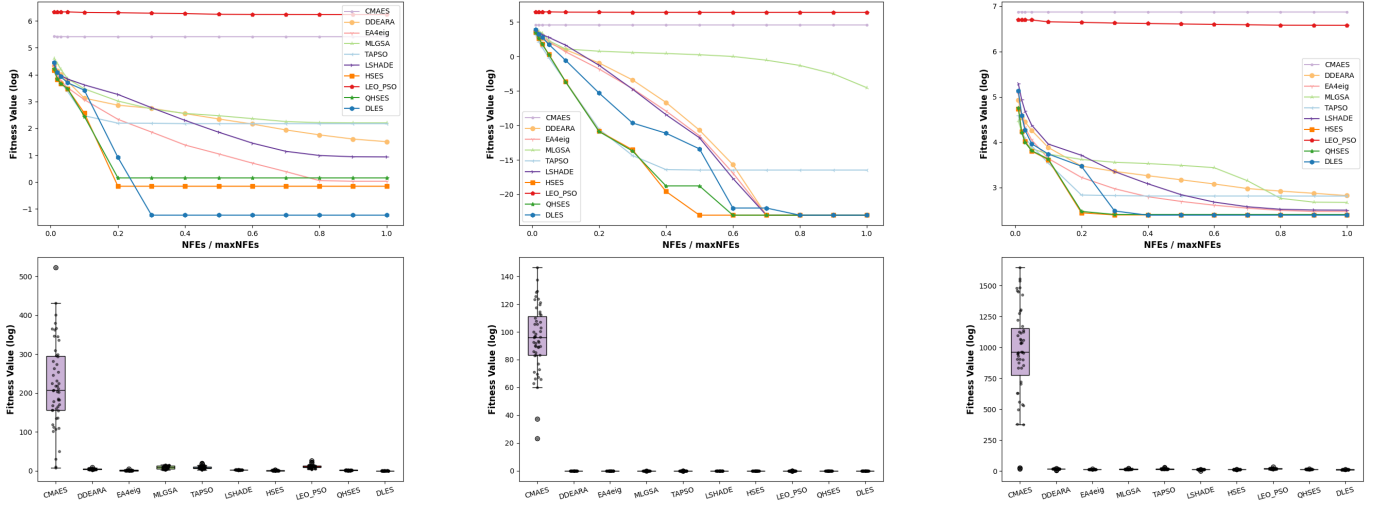


Fig. S.LI: F4 to F6 in D=10.

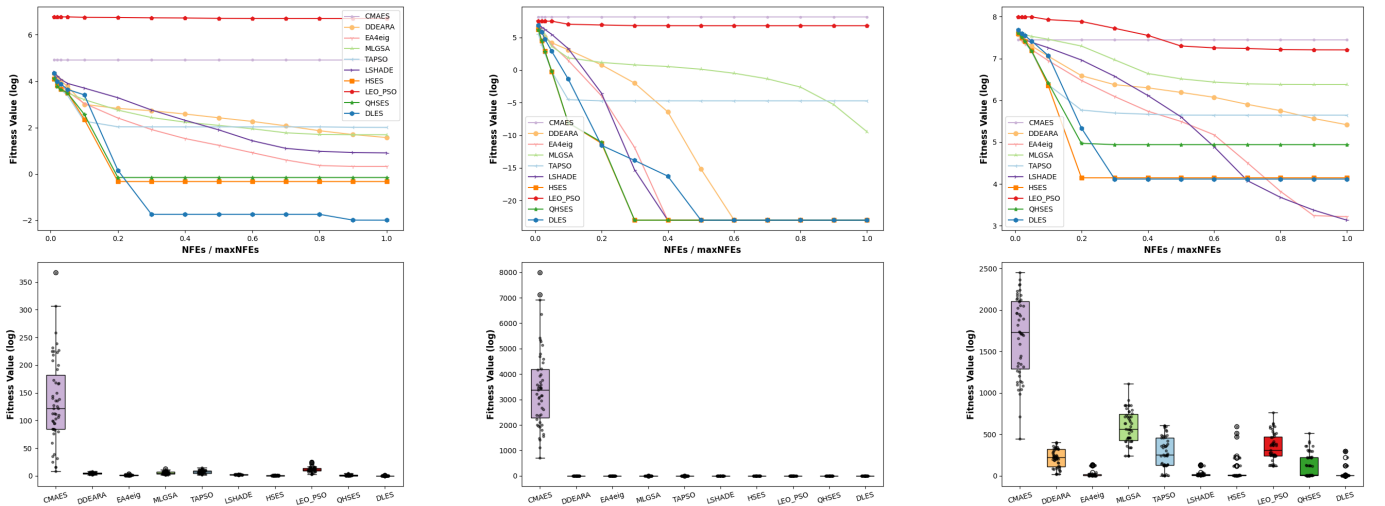


Fig. S.LII: F7 to F9 in D=10.

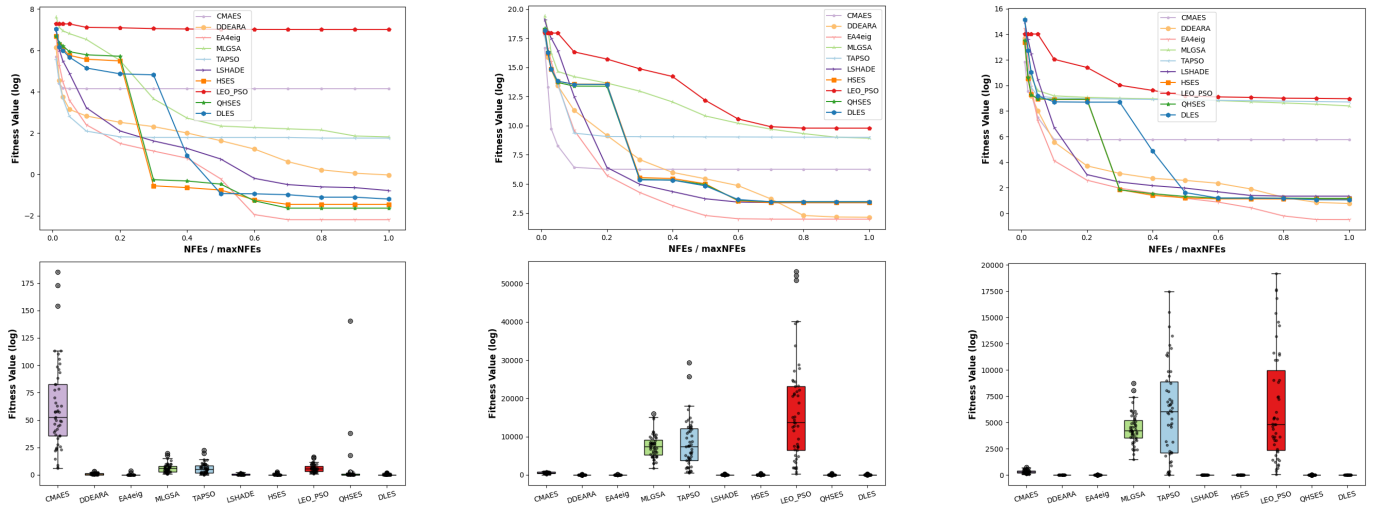


Fig. S.LIII: F10 to F12 in D=10.

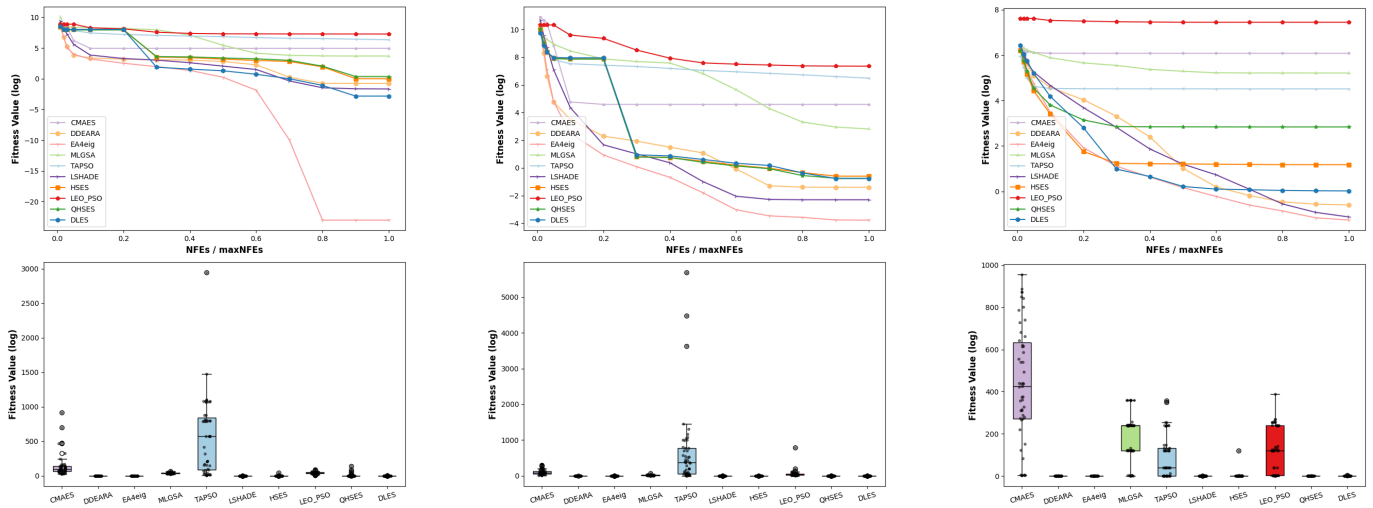


Fig. S.LIV: F13 to F15 in D=10.

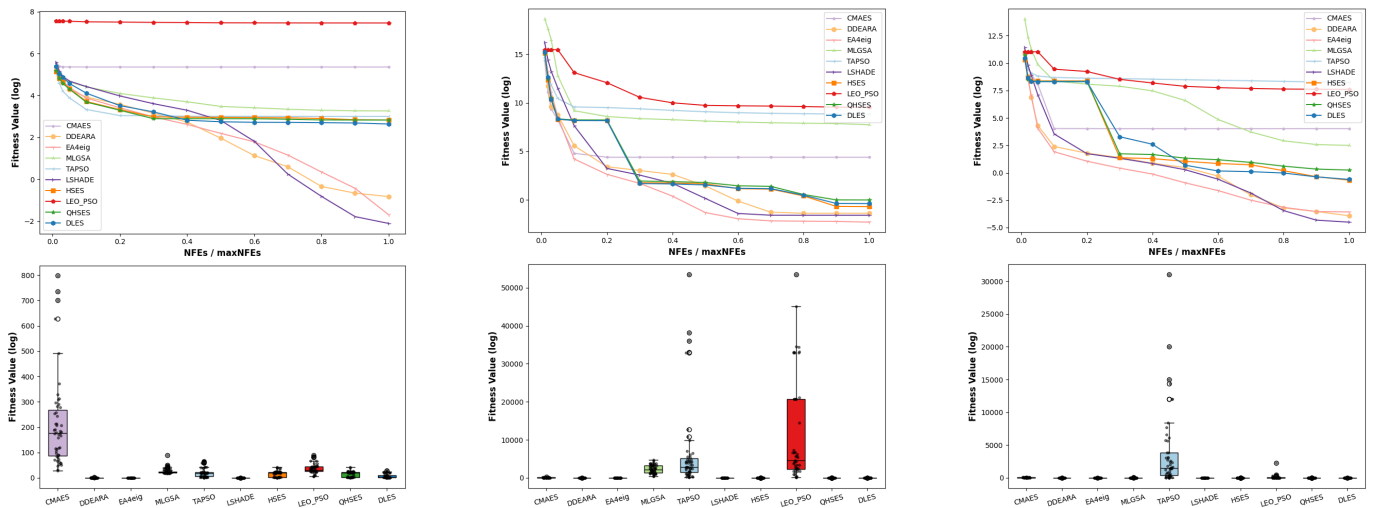


Fig. S.LV: F16 to F18 in D=10.

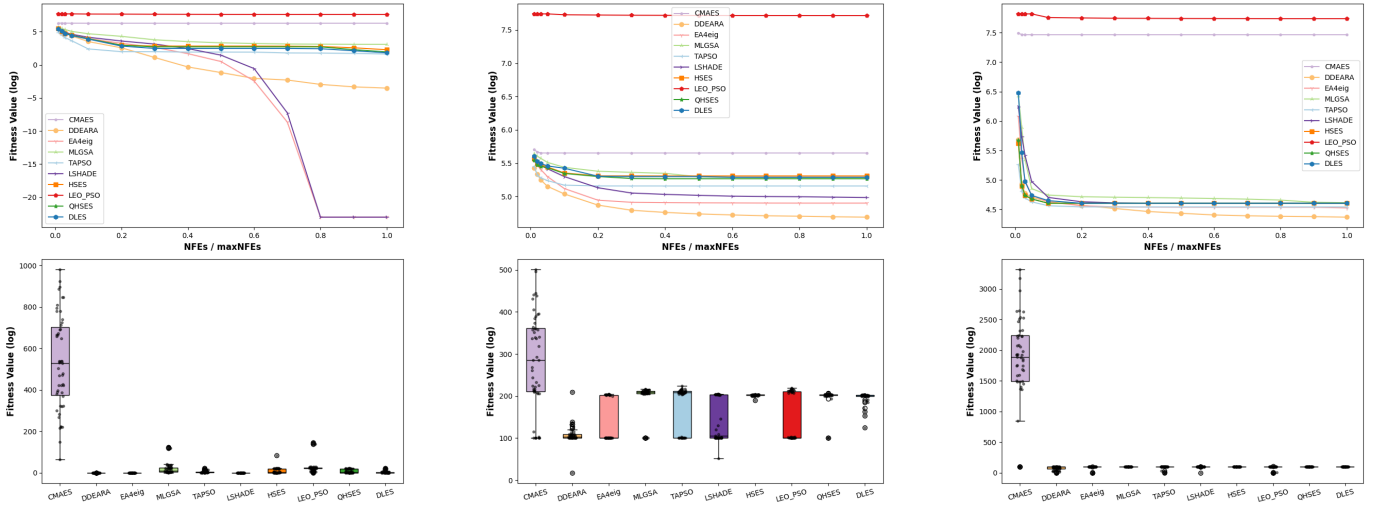


Fig. S.LVI: F19 to F21 in D=10.

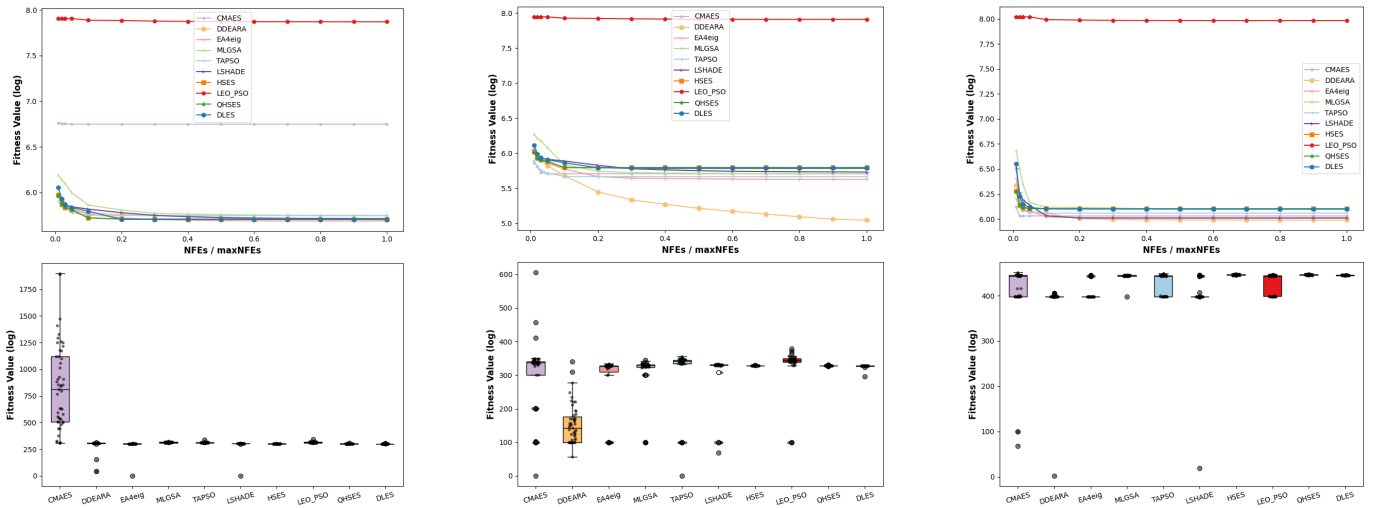


Fig. S.LVII: F22 to F24 in D=10.

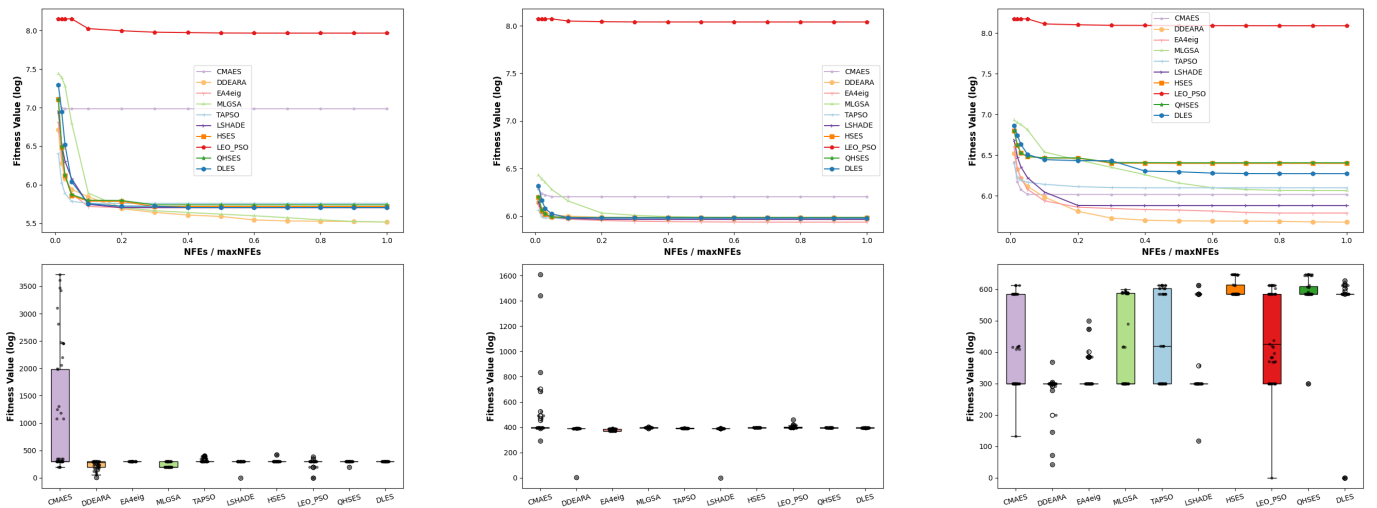


Fig. S.LVIII: F25 to F27 in D=10.

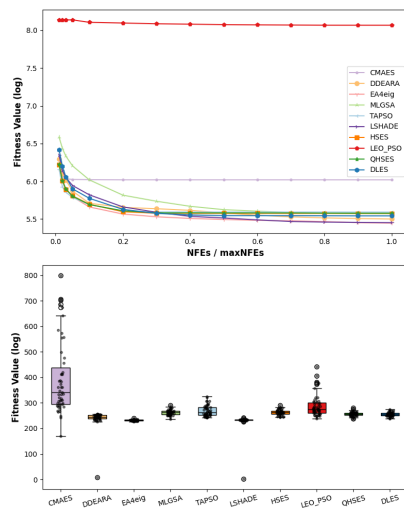


Fig. S.LIX: F28 and F29 in D=10.

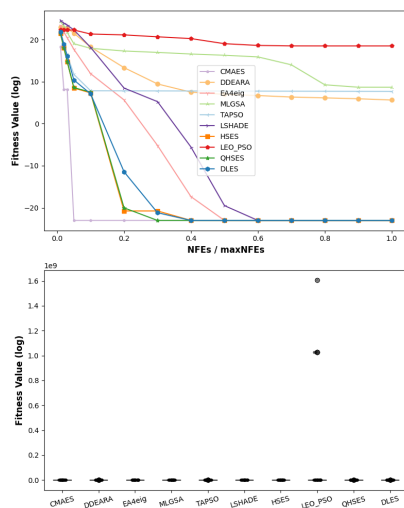
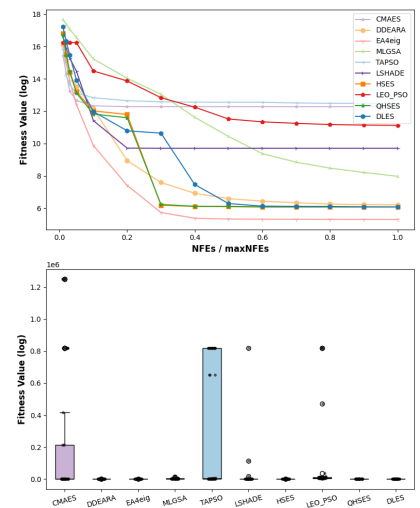


Fig. S.LX: F1 to F3 in D=30.

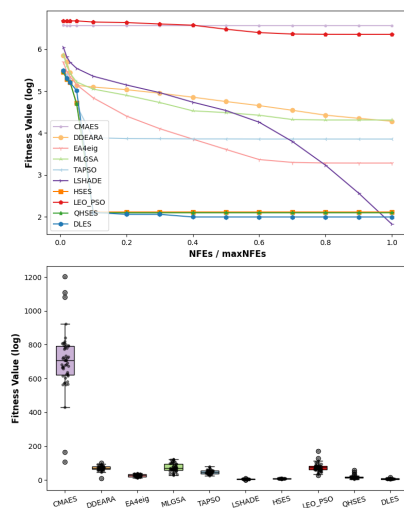
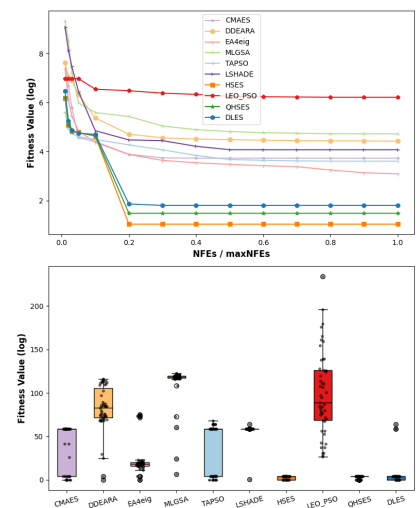
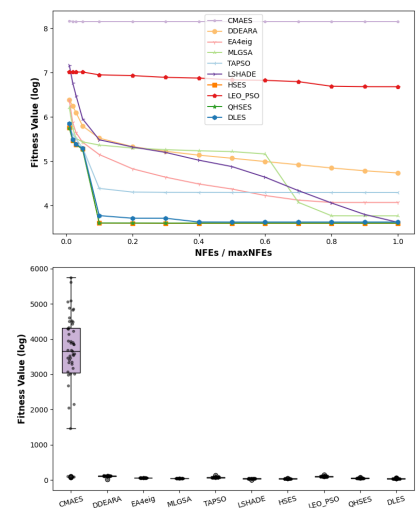


Fig. S.LXI: F4 to F6 in D=30.



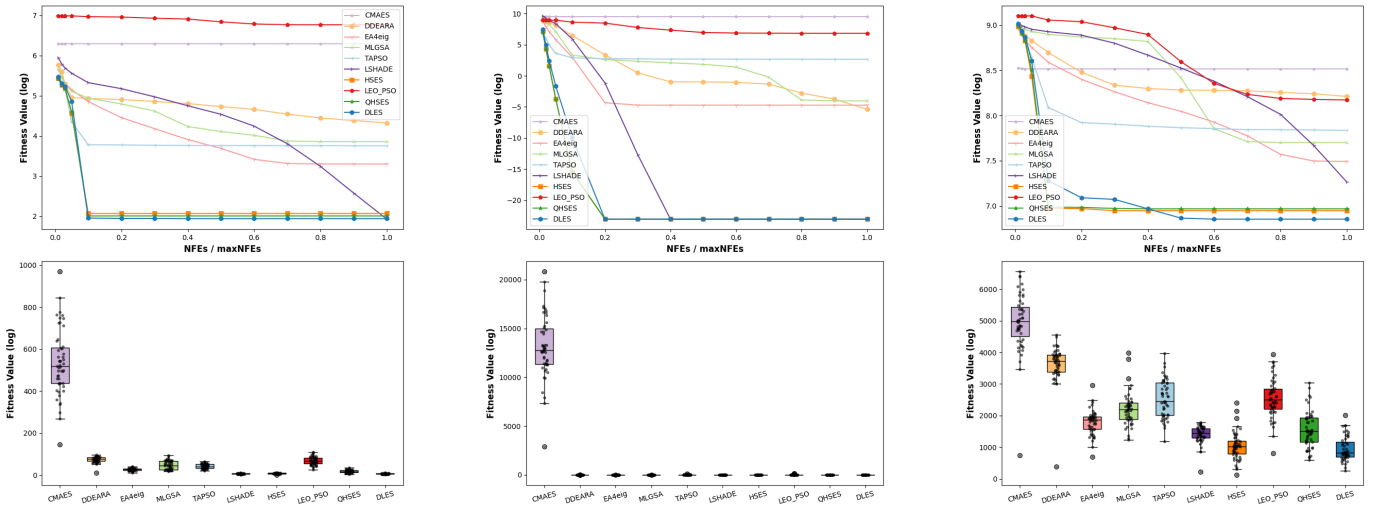


Fig. S.LXII: F7 to F9 in D=30.

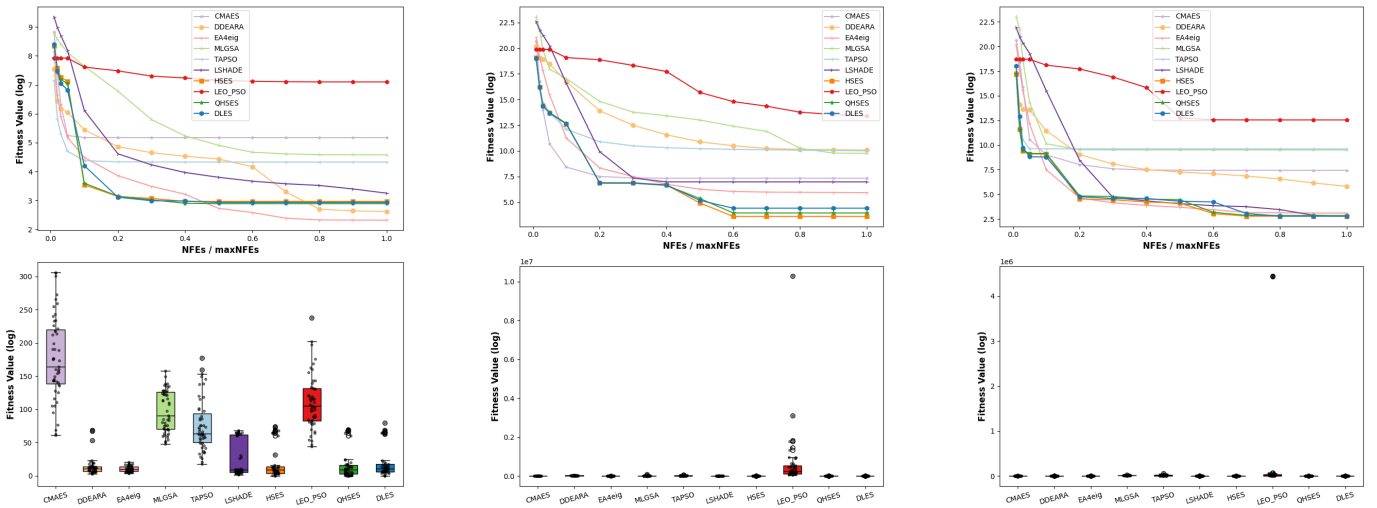


Fig. S.LXIII: F30 to F12 in D=30.

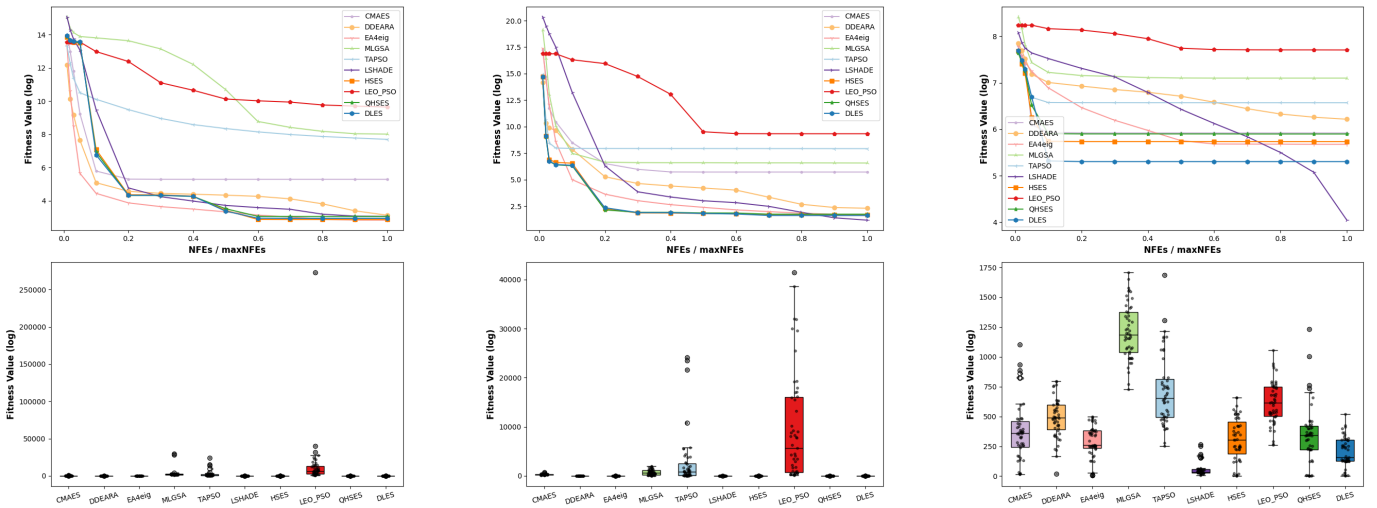


Fig. S.LXIV: F13 to F15 in D=30.

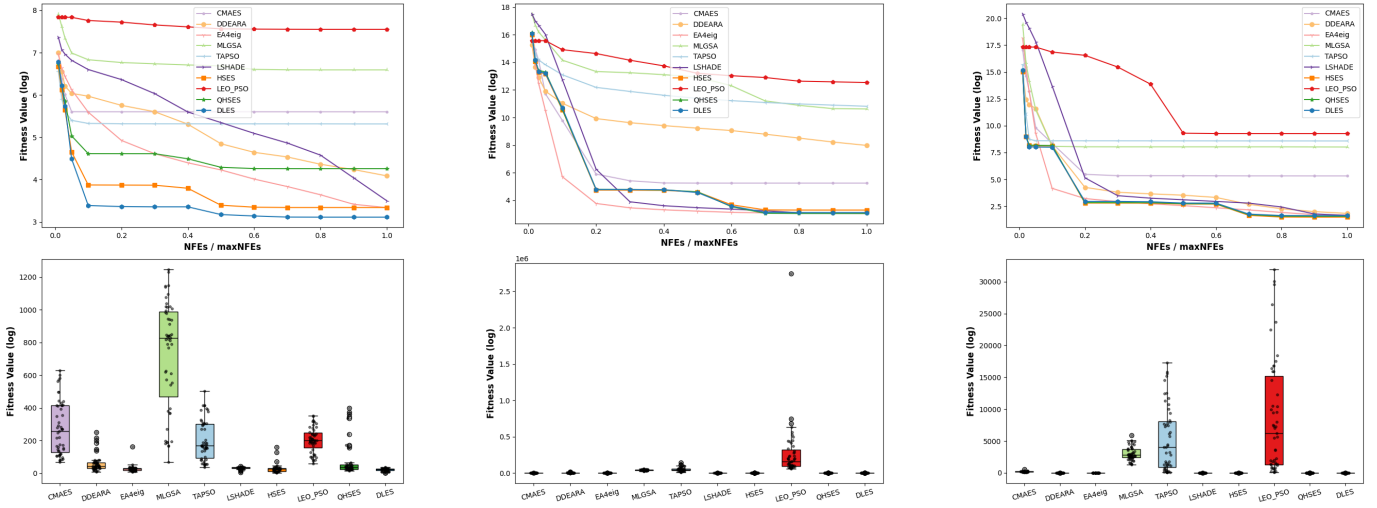


Fig. S.LXV: F16 to F18 in D=30.

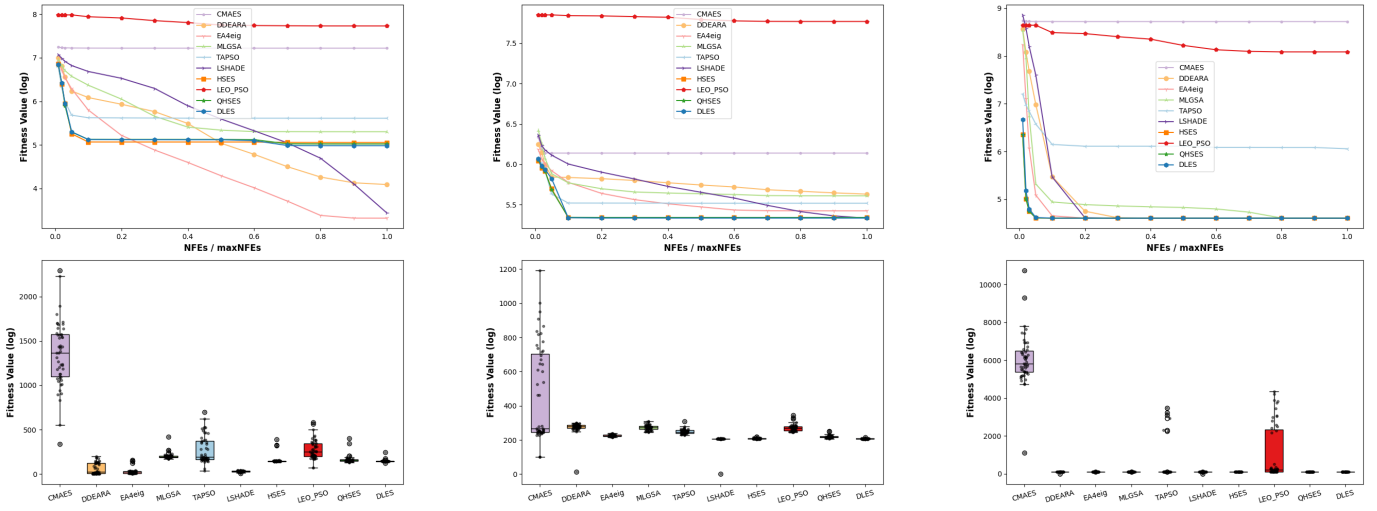


Fig. S.LXVI: F19 to F21 in D=30.

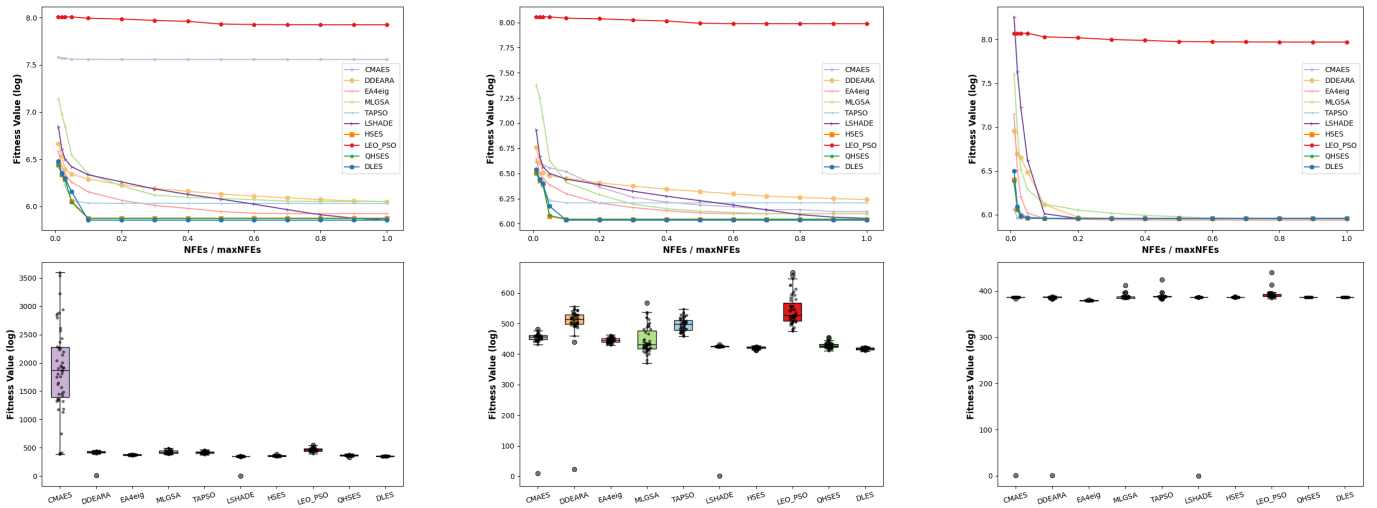


Fig. S.LXVII: F22 to F24 in D=30.

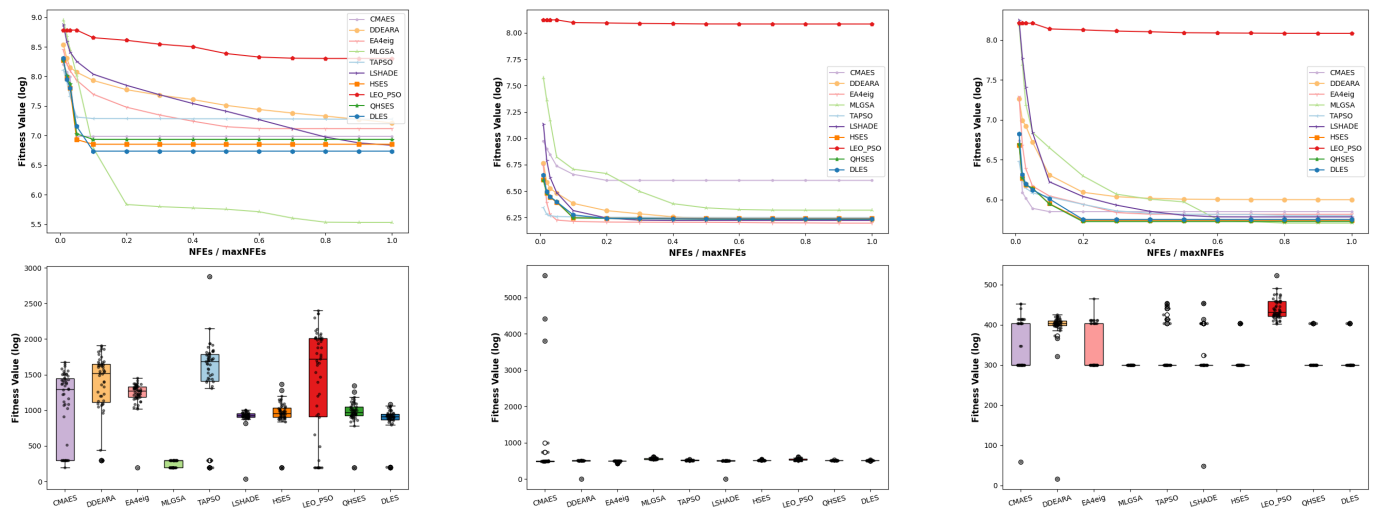


Fig. S.LXVIII: F25 to F27 in D=30.



Fig. S.LXIX: F28 and F29 in D=30.

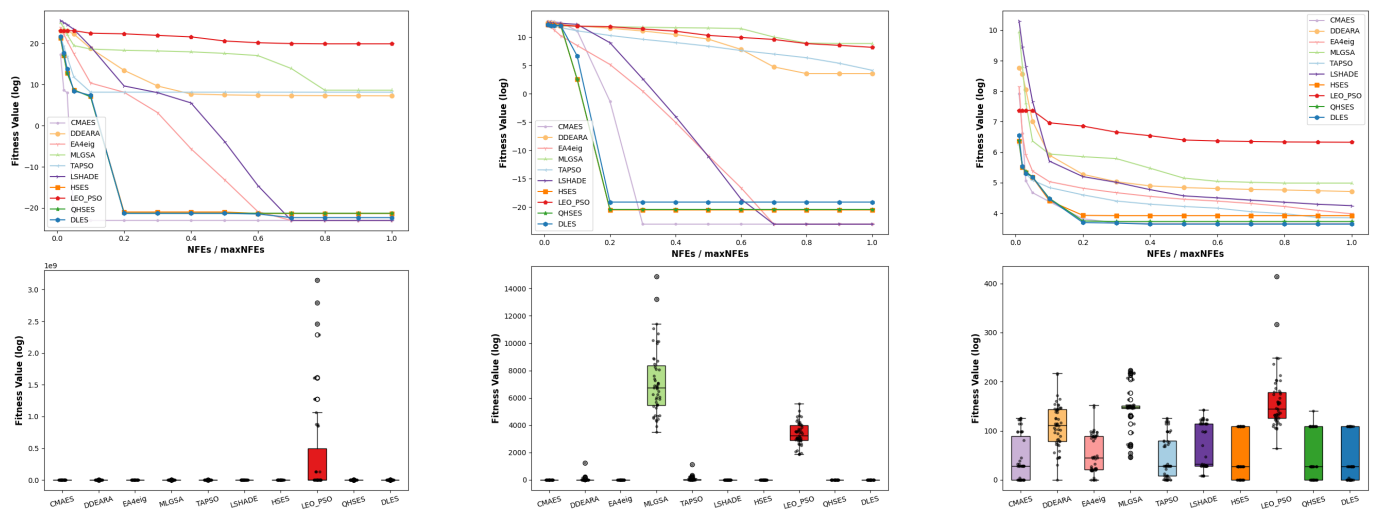


Fig. S.LXX: F1 to F3 in D=50.

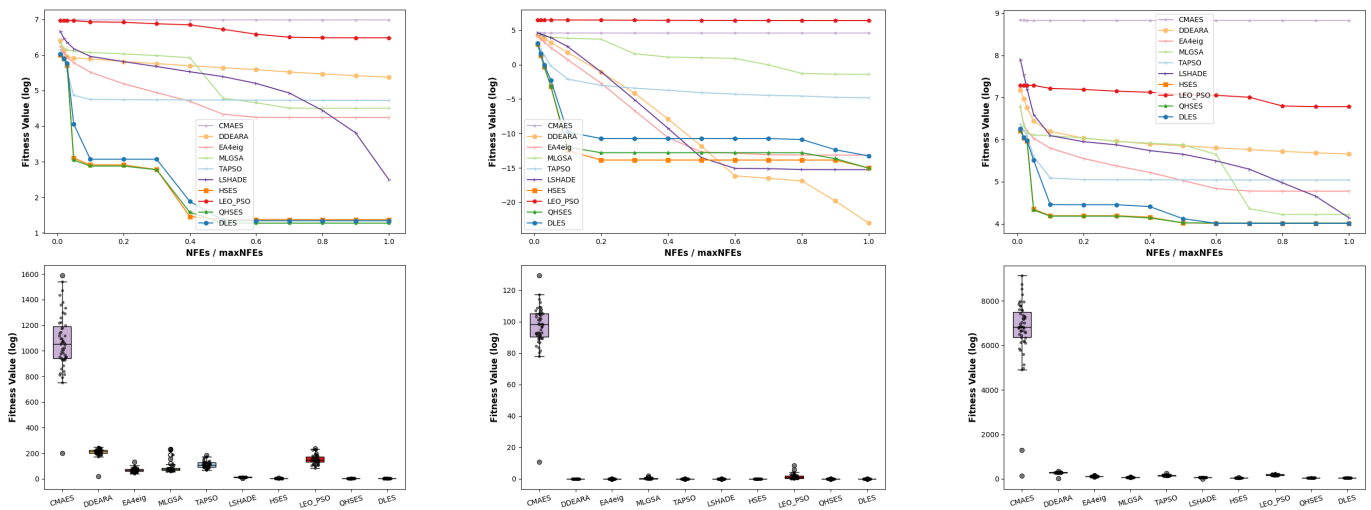


Fig. S.LXXI: F4 to F6 in D=50.

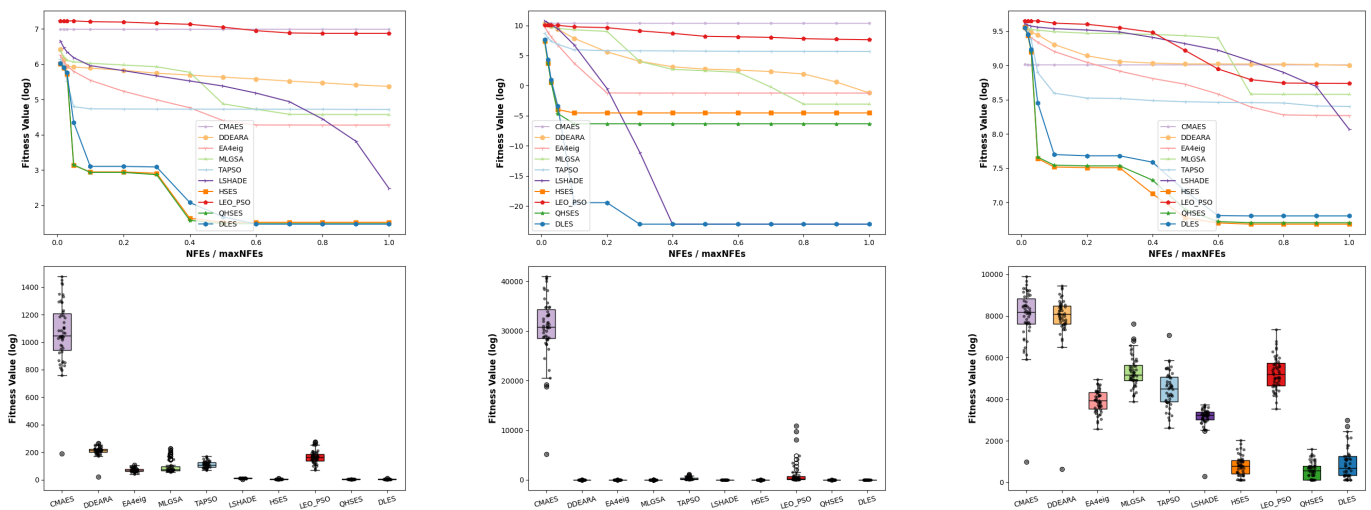


Fig. S.LXXII: F7 to F9 in D=50.

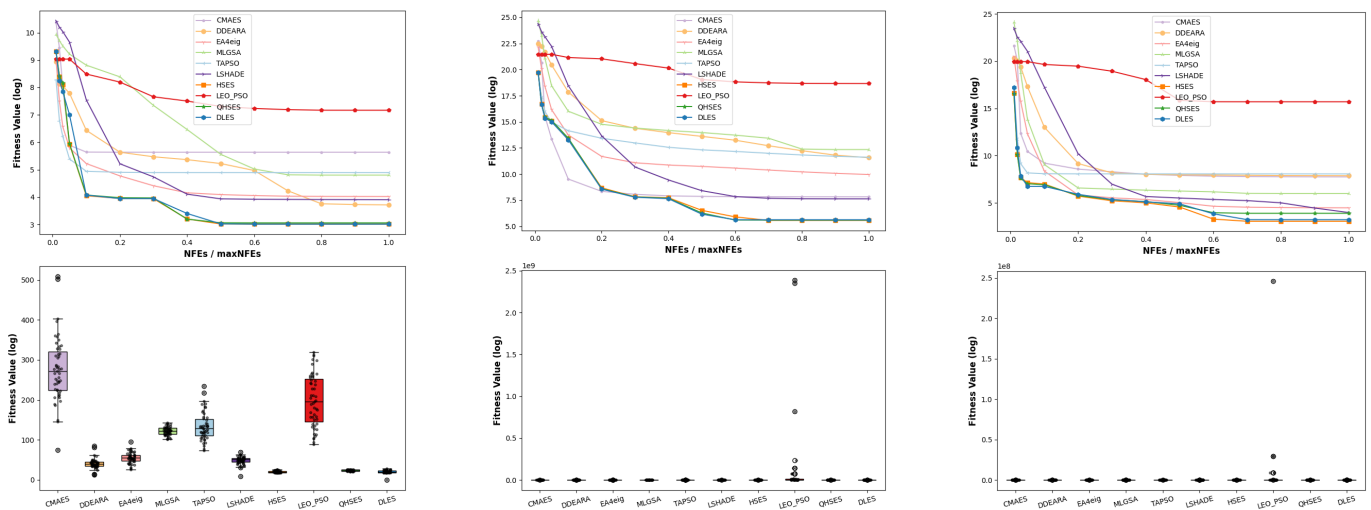


Fig. S.LXXIII: F50 to F12 in D=50.

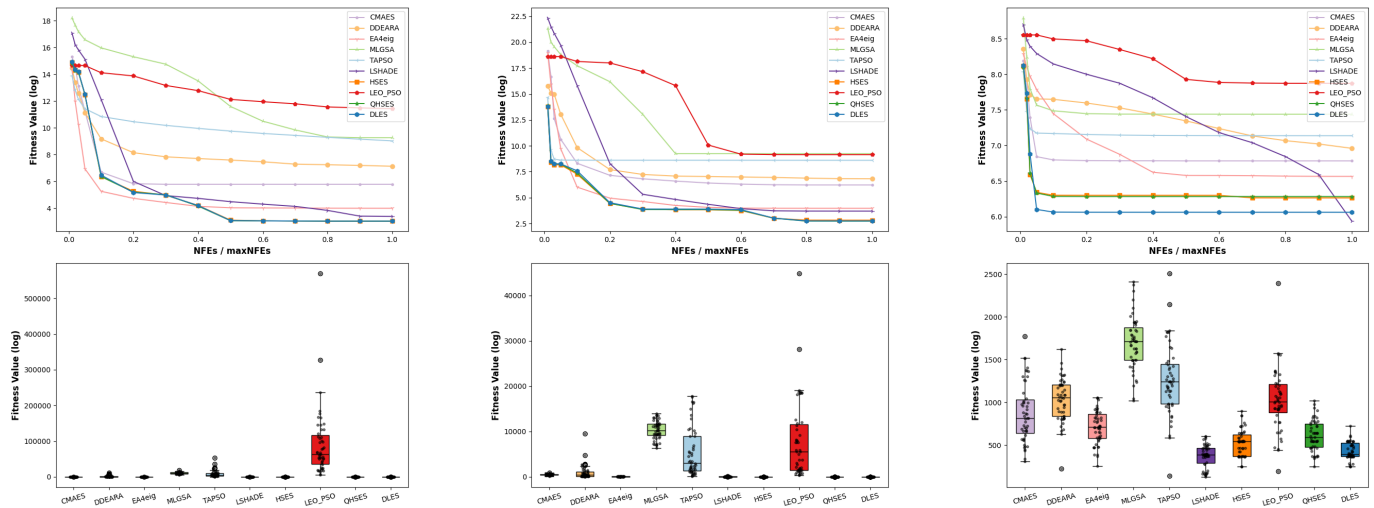


Fig. S.LXXIV: F13 to F15 in D=50.

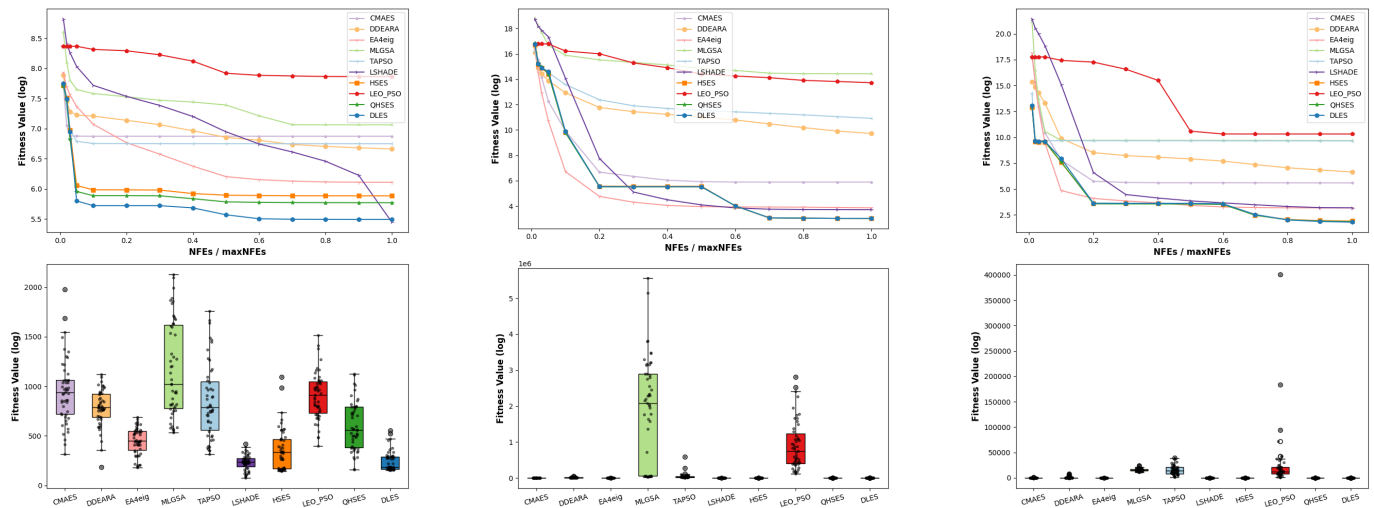


Fig. S.LXXV: F16 to F18 in D=50.

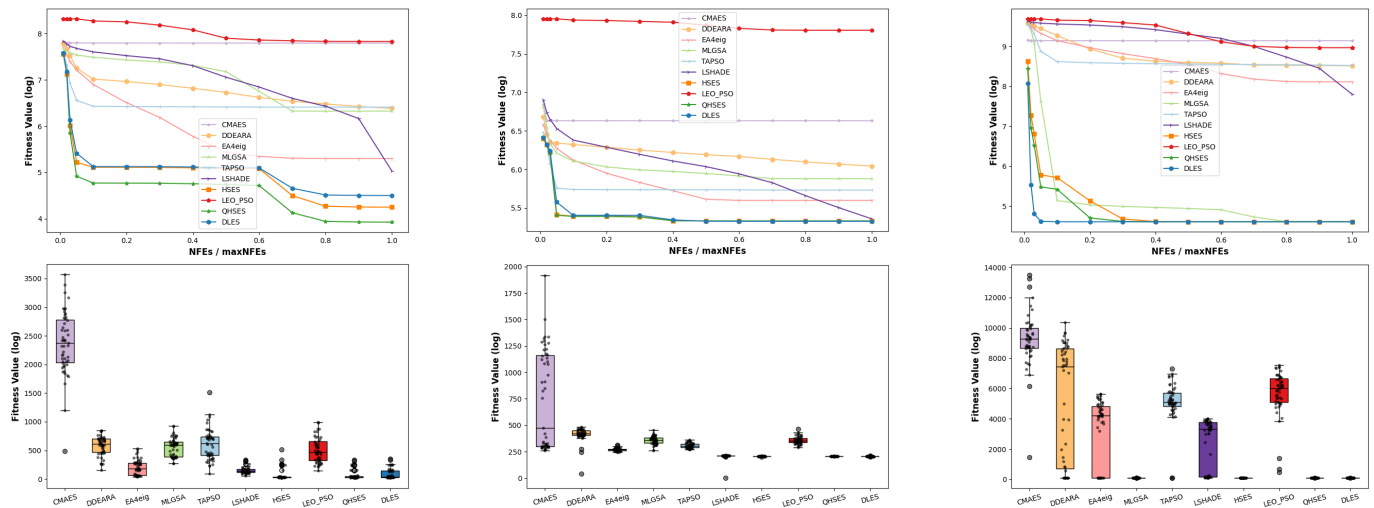


Fig. S.LXXVI: F19 to F21 in D=50.

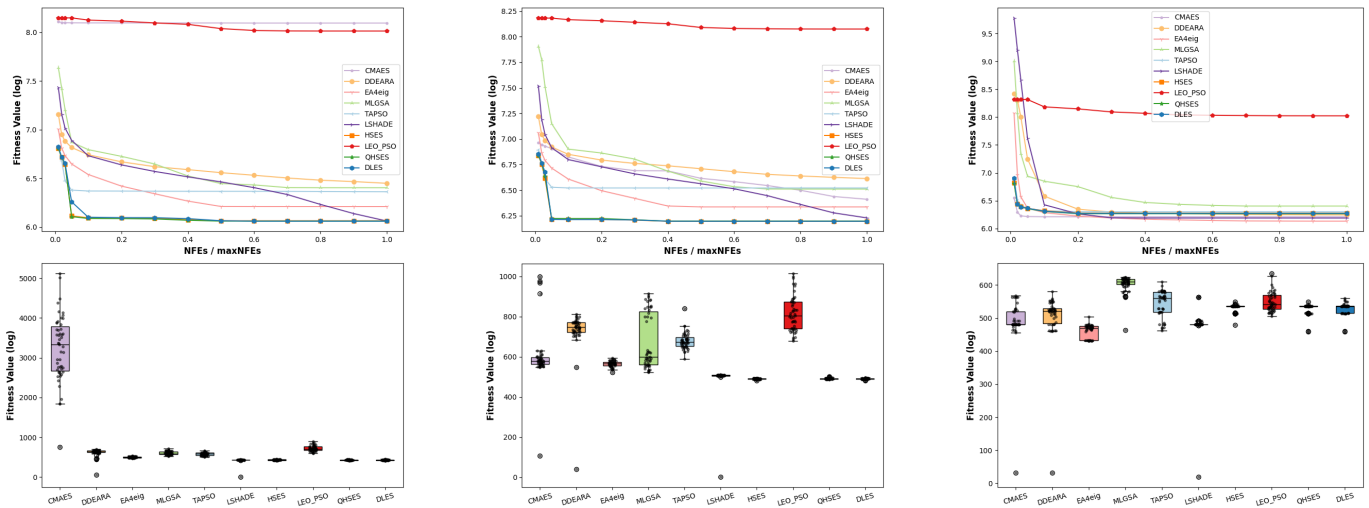


Fig. S.LXXVII: F22 to F24 in D=50.

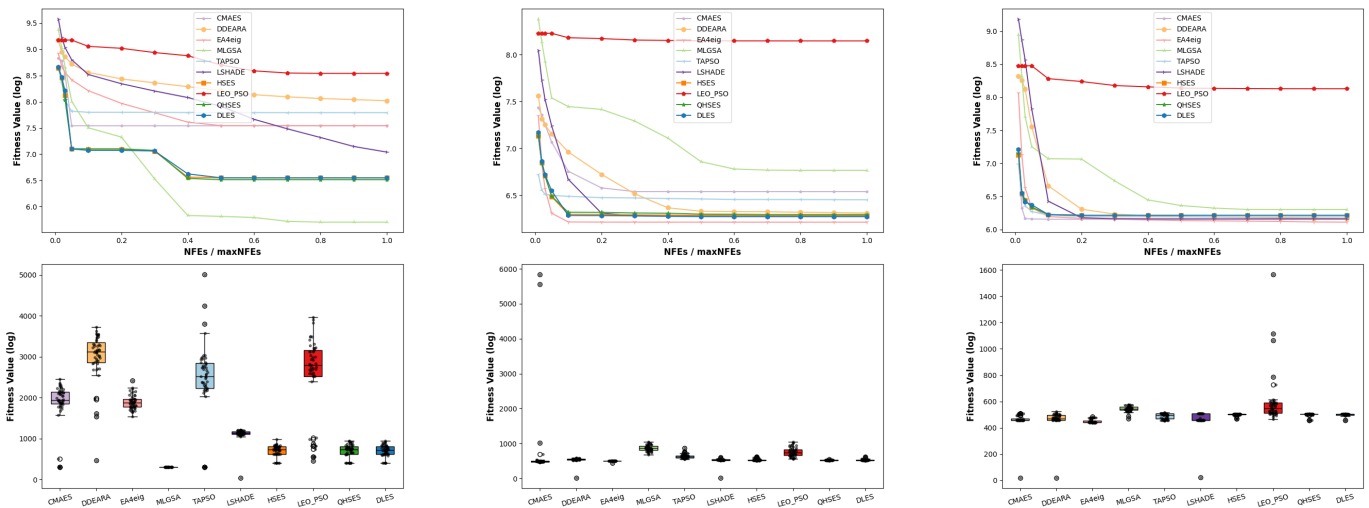


Fig. S.LXXVIII: F25 to F27 in D=50.



Fig. S.LXXIX: F28 and F29 in D=50.

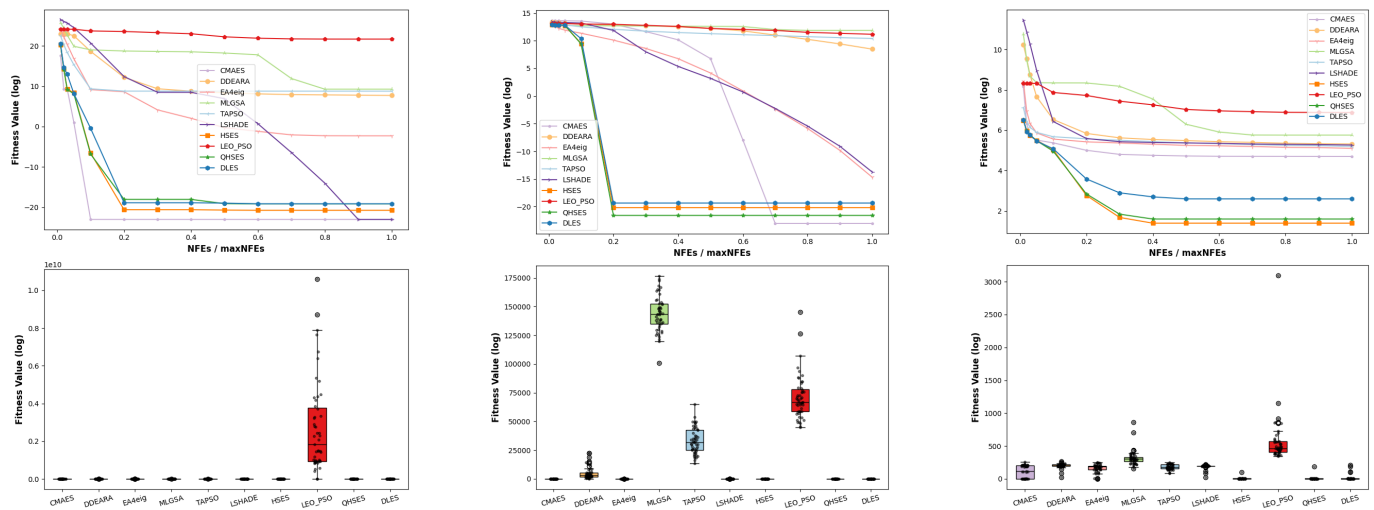


Fig. S.LXXX: F1 to F3 in D=100.

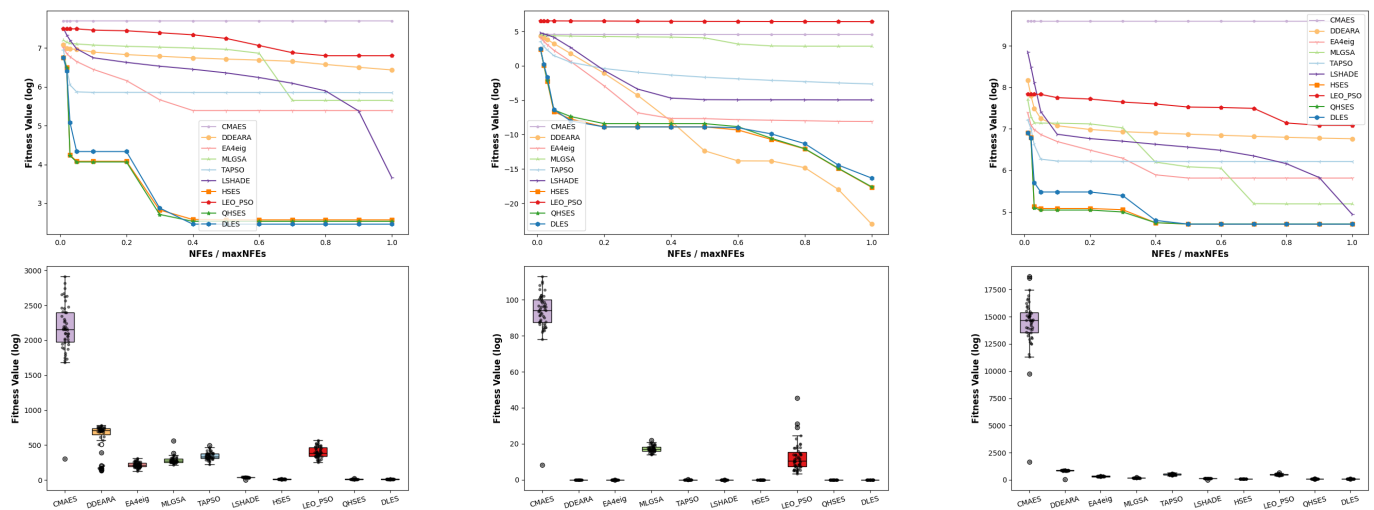


Fig. S.LXXXI: F4 to F6 in D=100.

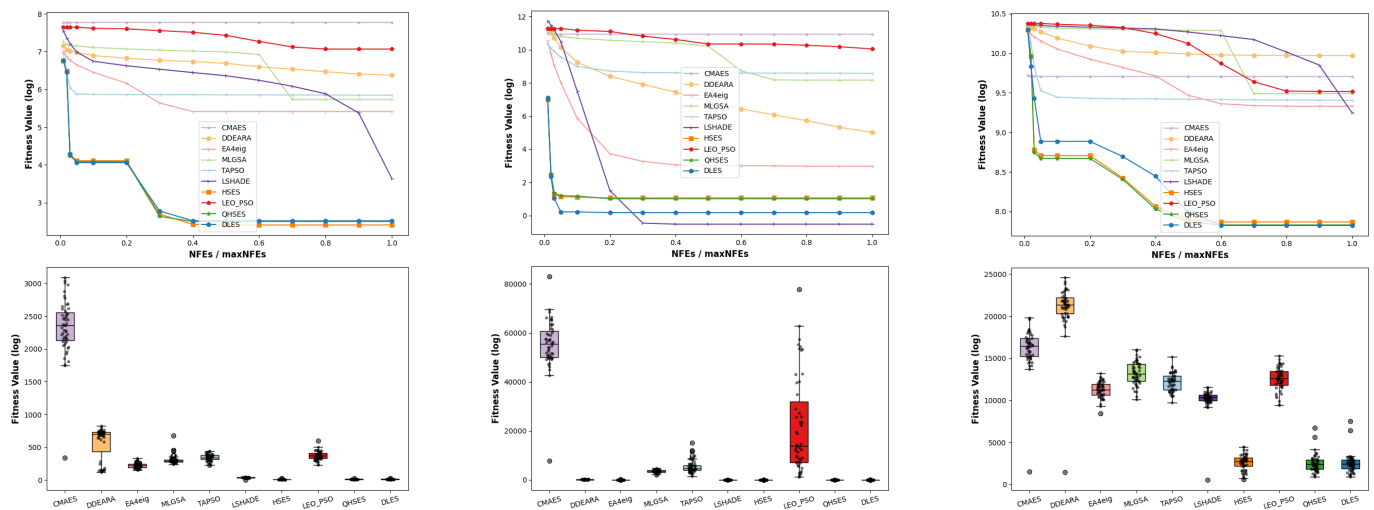


Fig. S.LXXXII: F7 to F9 in D=100.

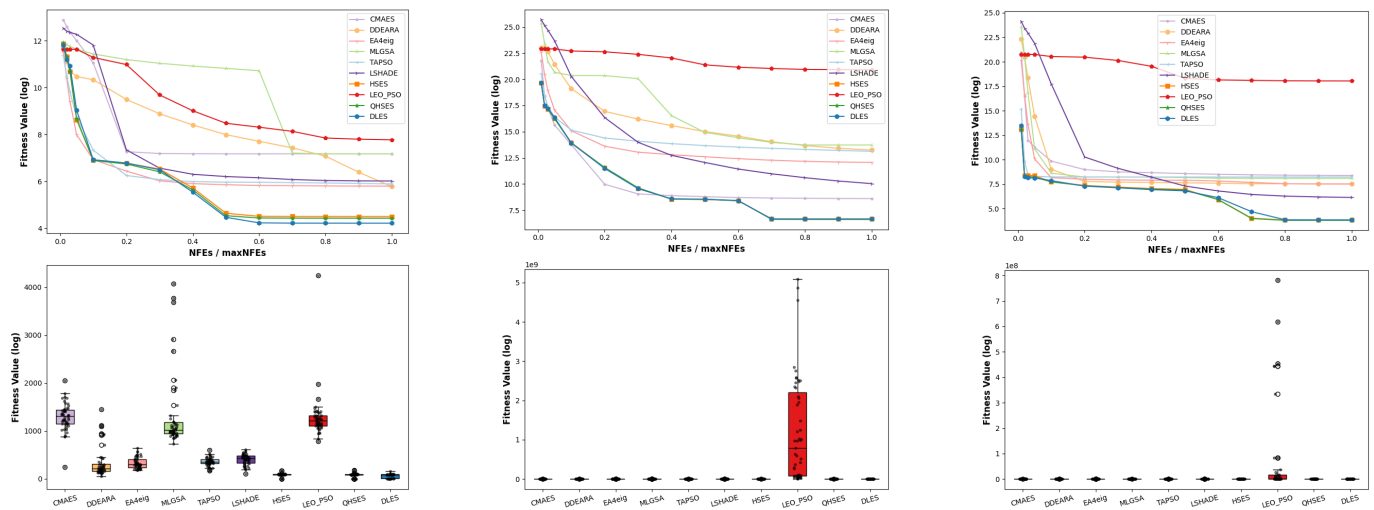


Fig. S.LXXXIII: F100 to F12 in D=100.

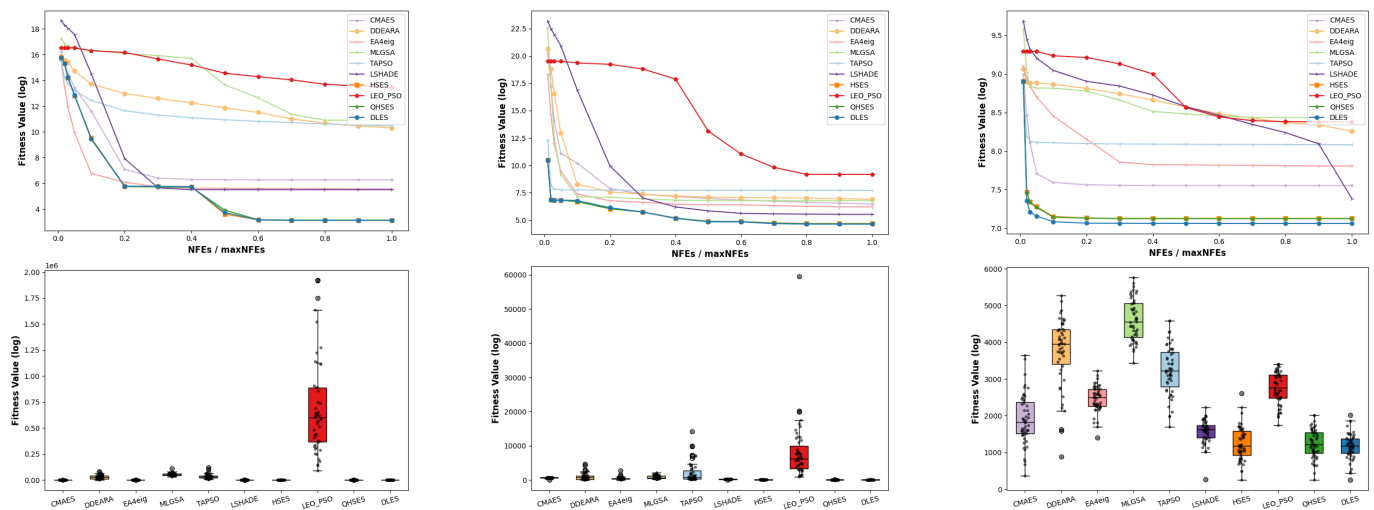


Fig. S.LXXXIV: F13 to F15 in D=100.

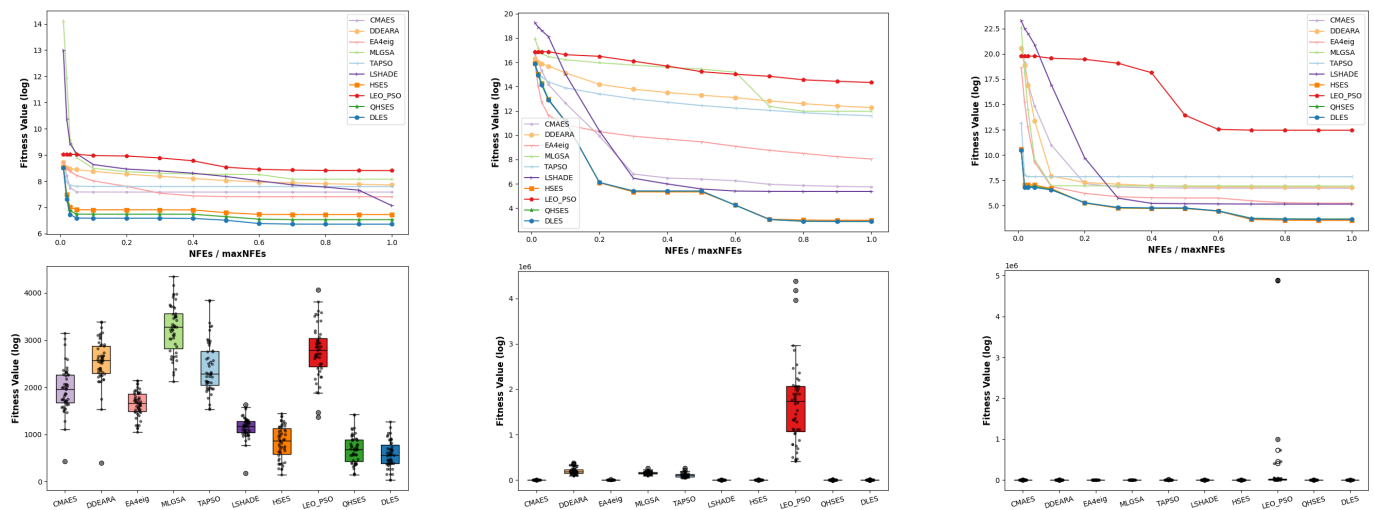


Fig. S.LXXXV: F16 to F18 in D=100.

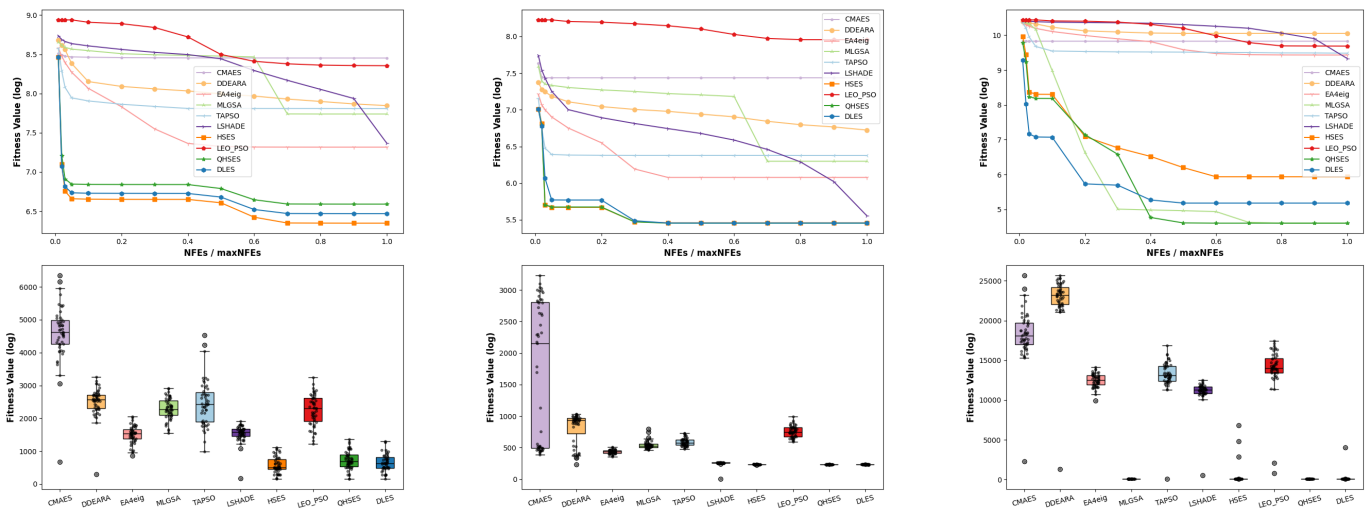


Fig. S.LXXXVI: F19 to F21 in D=100.

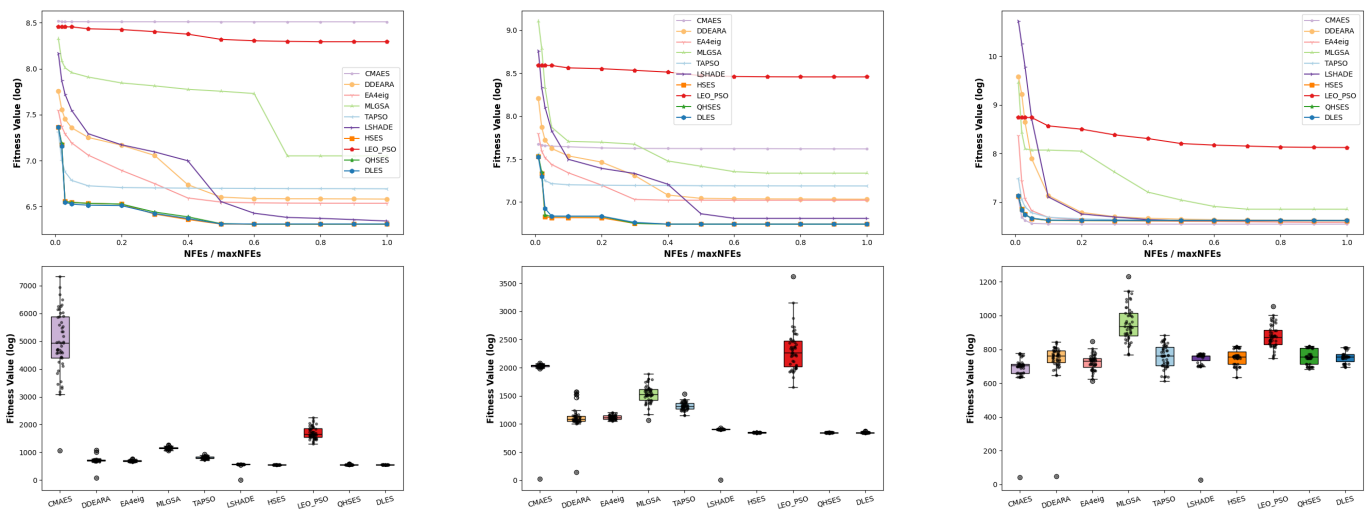


Fig. S.LXXXVII: F22 to F24 in D=100.

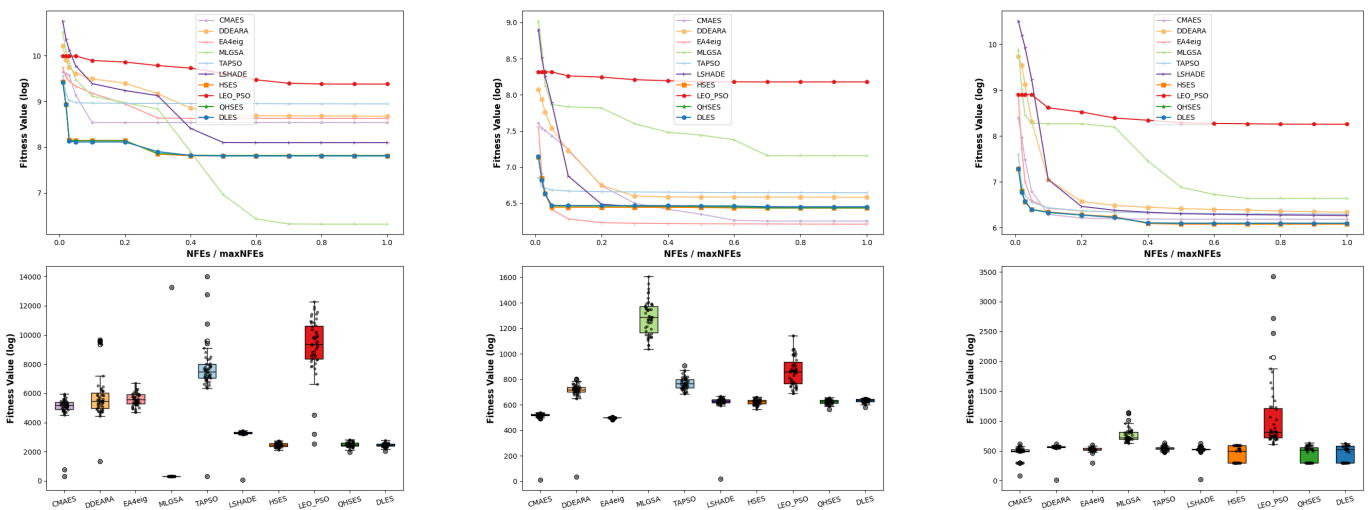


Fig. S.LXXXVIII: F25 to F27 in D=100.



Fig. S.LXXXIX: F28 and F29 in D=100.